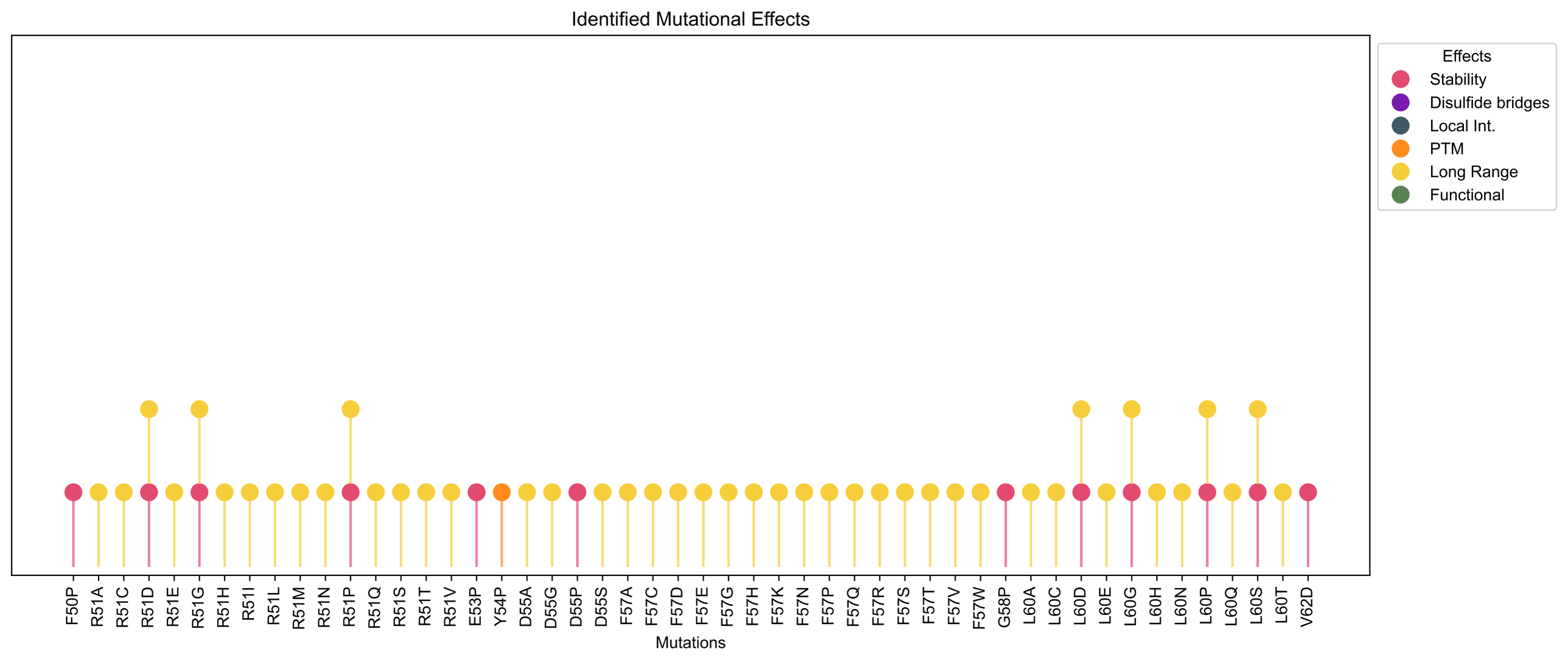
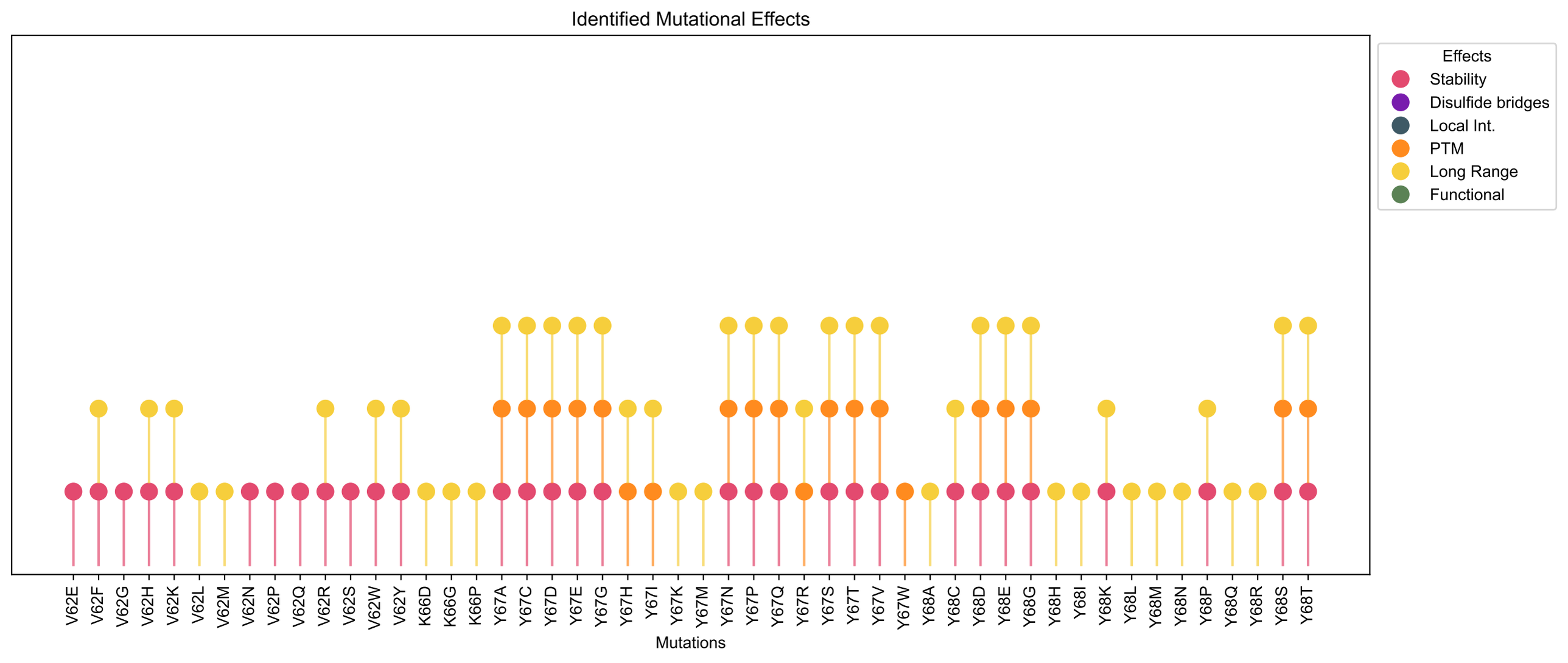
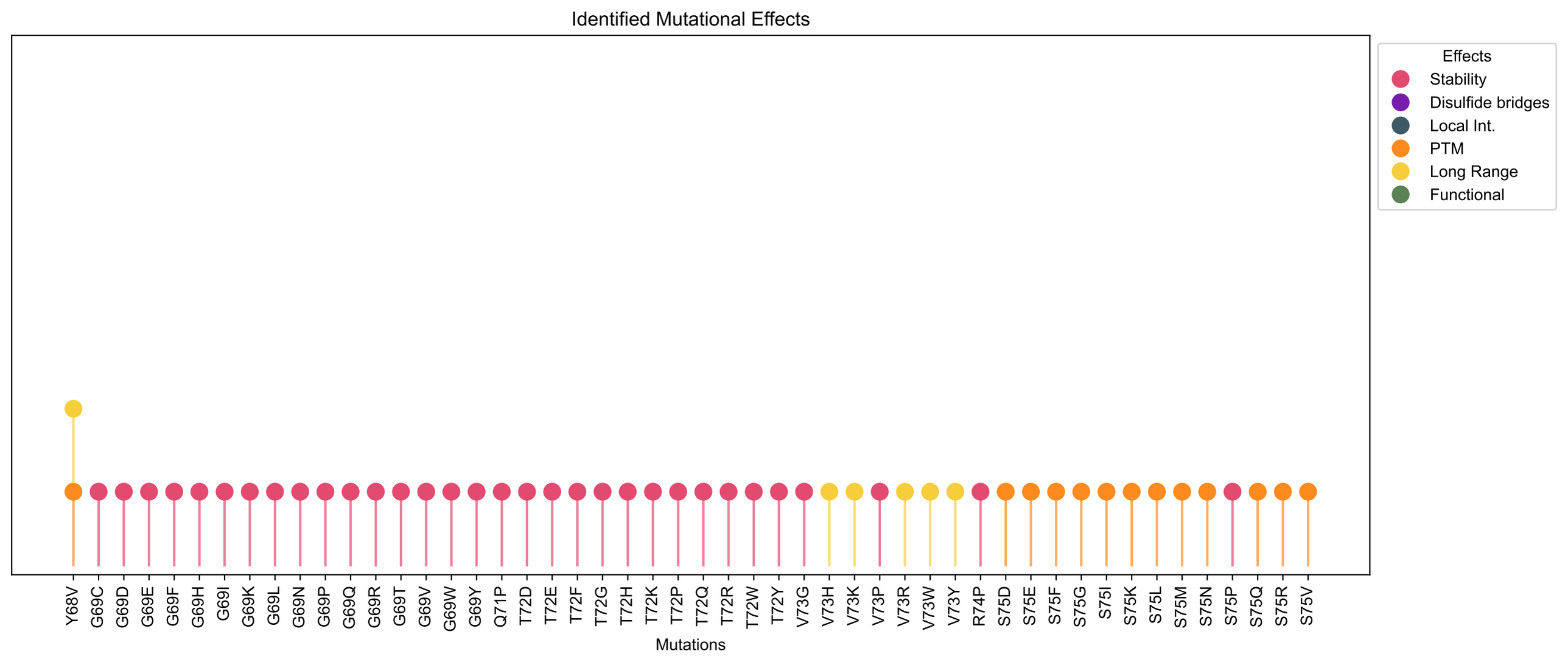
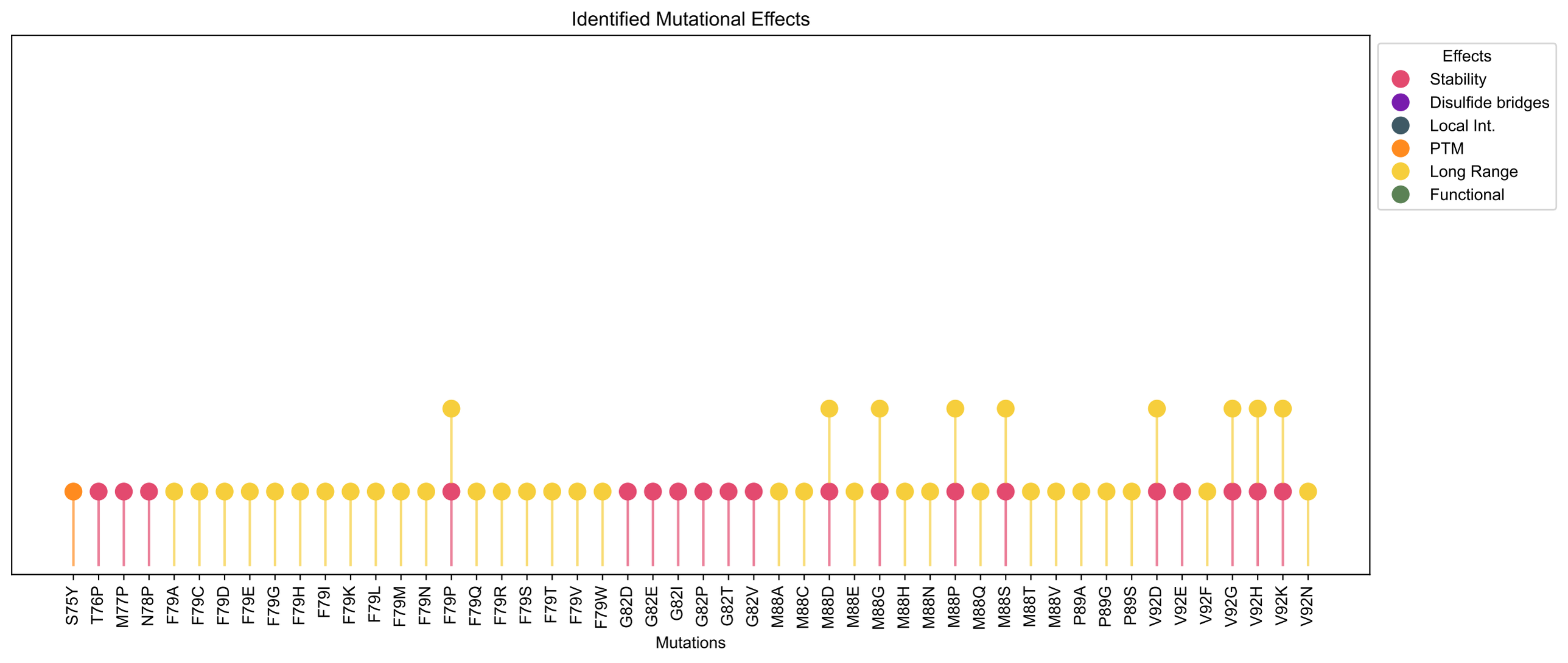
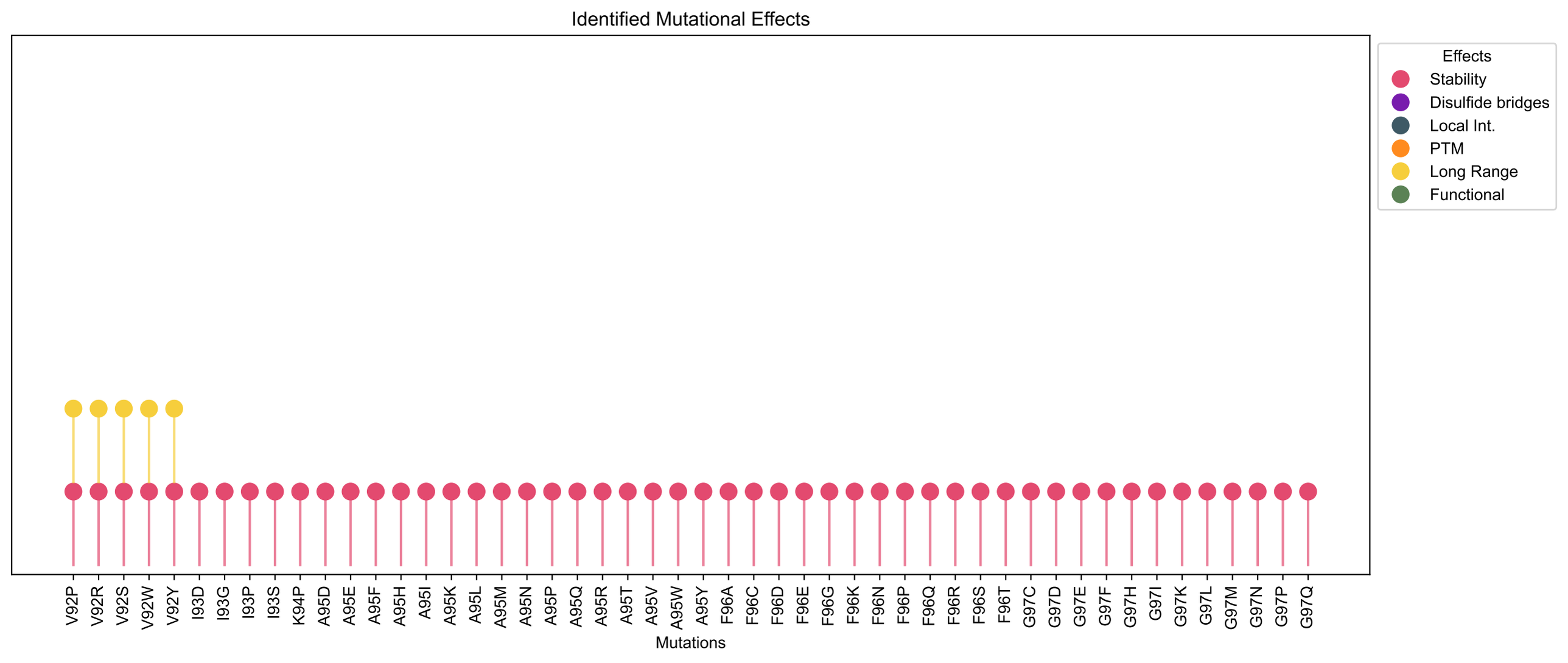
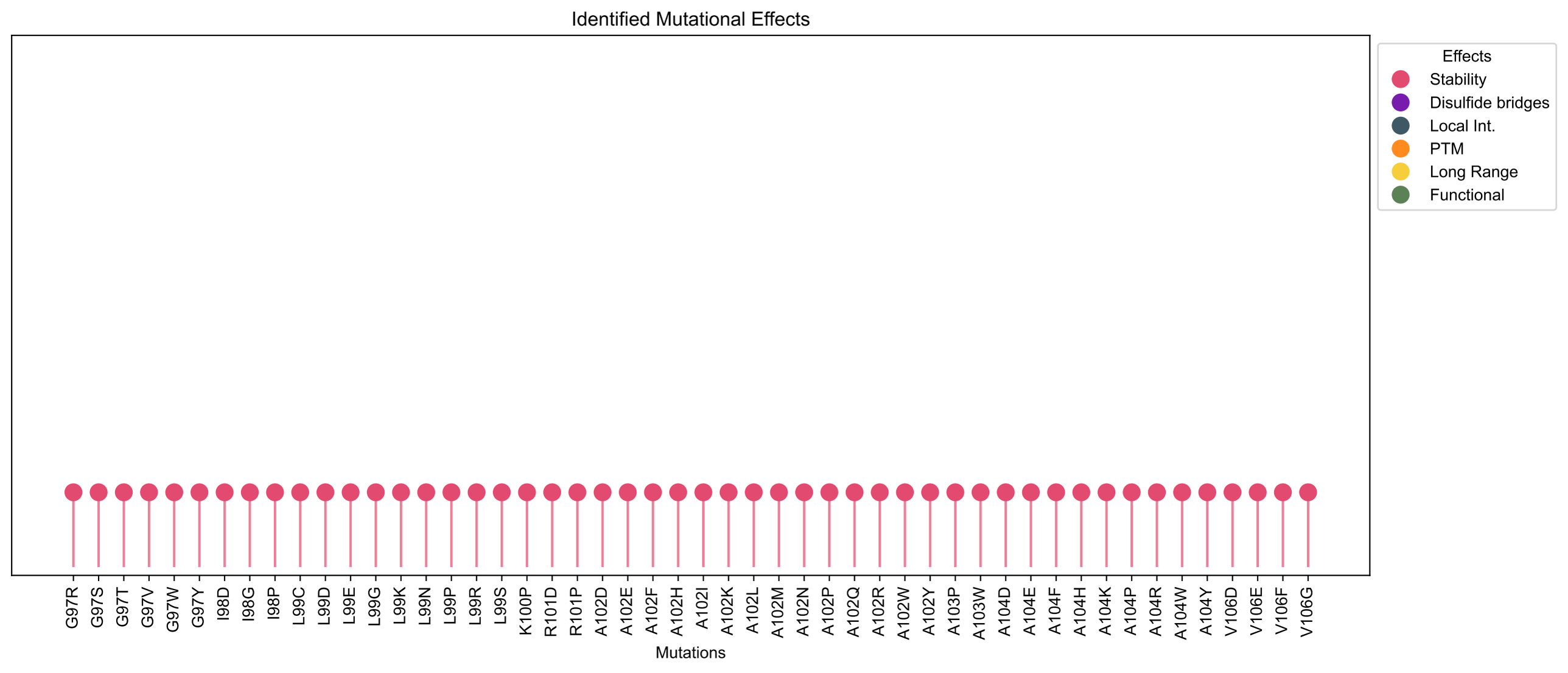

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
F50P
R51A
R51C
R51D
R51E
R51G
R51H
R51I
R51L
R51M
R51N
R51P
R51Q
R51S
R51T
R51V
E53P
Y54P
D55A
D55G
D55P
D55S
F57A
F57C
F57D
F57E
F57G
F57H
F57K
F57N
F57P
F57Q
F57R
F57S
F57T
F57V
F57W
G58P
L60A
L60C
L60D
L60E
L60G
L60H
L60N
L60P
L60Q
L60S
L60T
V62D
Mutations

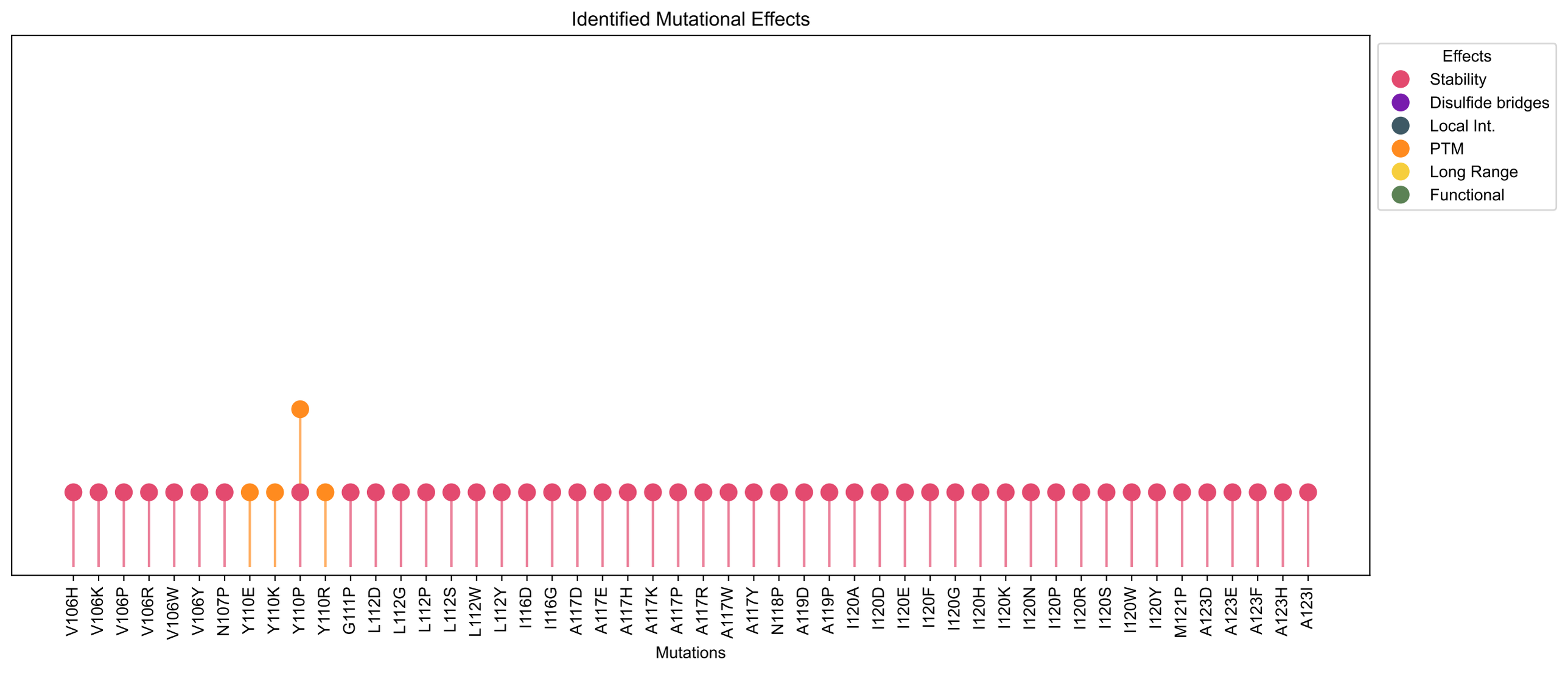

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
V62E
V62F
V62G
V62H
V62K
V62L
V62M
V62N
V62P
V62Q
V62R
V62S
V62W
V62Y
K66D
K66G
K66P
Y67A
Y67C
Y67D
Y67E
Y67G
Y67H
Y67I
Y67K
Y67M
Y67N
Y67P
Y67Q
Y67R
Y67S
Y67T
Y67V
Y67W
Y68A
Y68C
Y68D
Y68E
Y68G
Y68H
Y68I
Y68K
Y68L
Y68M
Y68N
Y68P
Y68Q
Y68R
Y68S
Y68T
Mutations

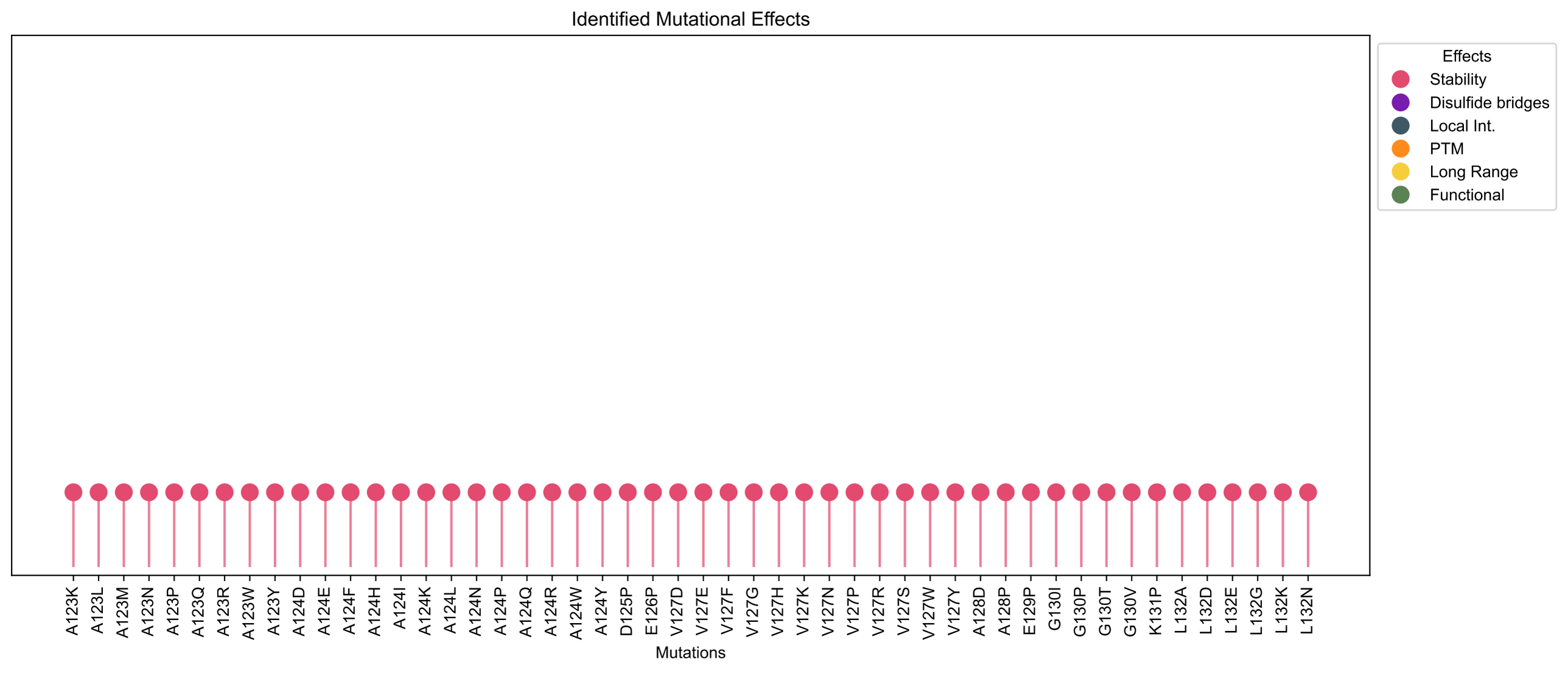

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
Y68V
G69C
G69D
G69E
G69F
G69H
G69I
G69K
G69L
G69N
G69P
G69Q
G69R
G69T
G69V
G69W
G69Y
Q71P
T72D
T72E
T72F
T72G
T72H
T72K
T72P
T72Q
T72R
T72W
T72Y
V73G
V73H
V73K
V73P
V73R
V73W
V73Y
R74P
S75D
S75E
S75F
S75G
S75I
S75K
S75L
S75M
S75N
S75P
S75Q
S75R
S75V
Mutations

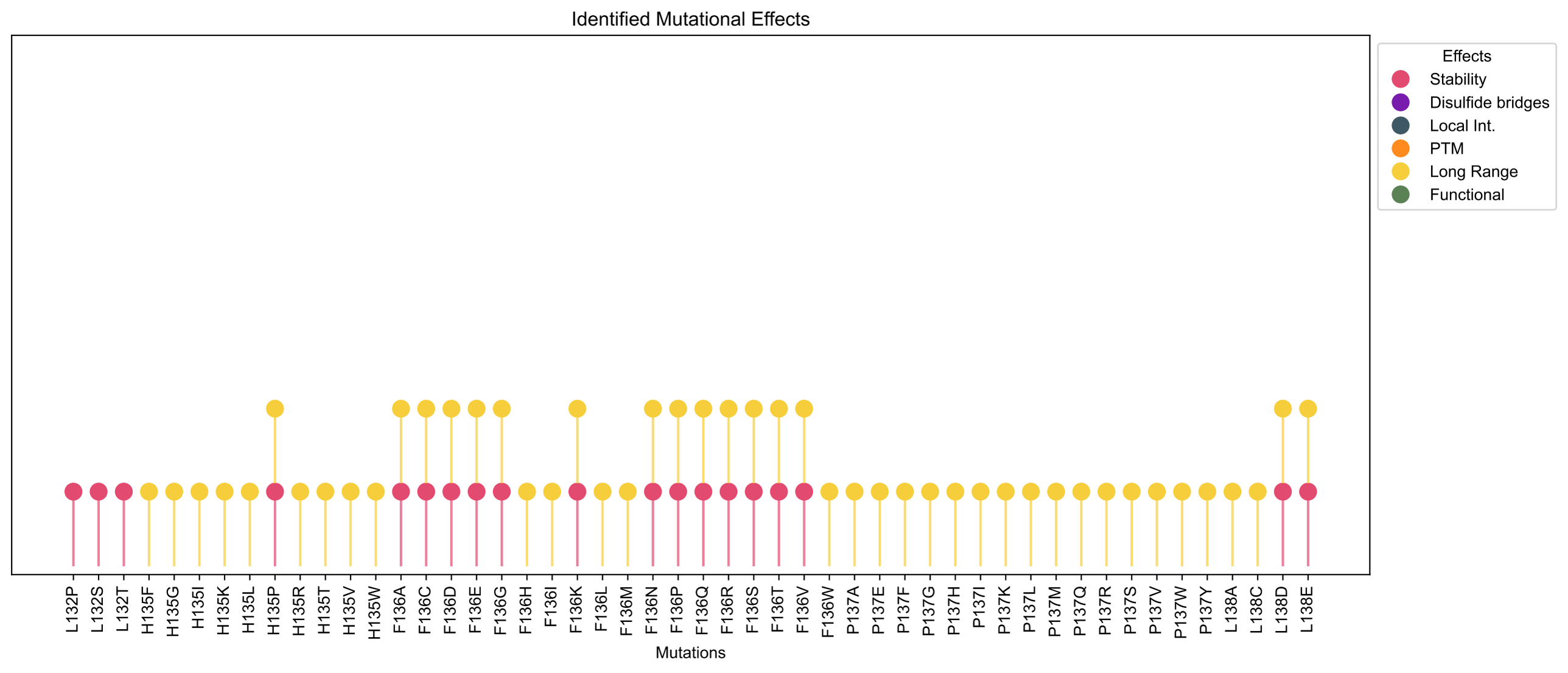

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
S75Y
T76P
M77P
N78P
F79A
F79C
F79D
F79E
F79G
F79H
F79I
F79K
F79L
F79M
F79N
F79P
F79Q
F79R
F79S
F79T
F79V
F79W
G82D
G82E
G82I
G82P
G82T
G82V
M88A
M88C
M88D
M88E
M88G
M88H
M88N
M88P
M88Q
M88S
M88T
M88V
P89A
P89G
P89S
V92D
V92E
V92F
V92G
V92H
V92K
V92N
Mutations

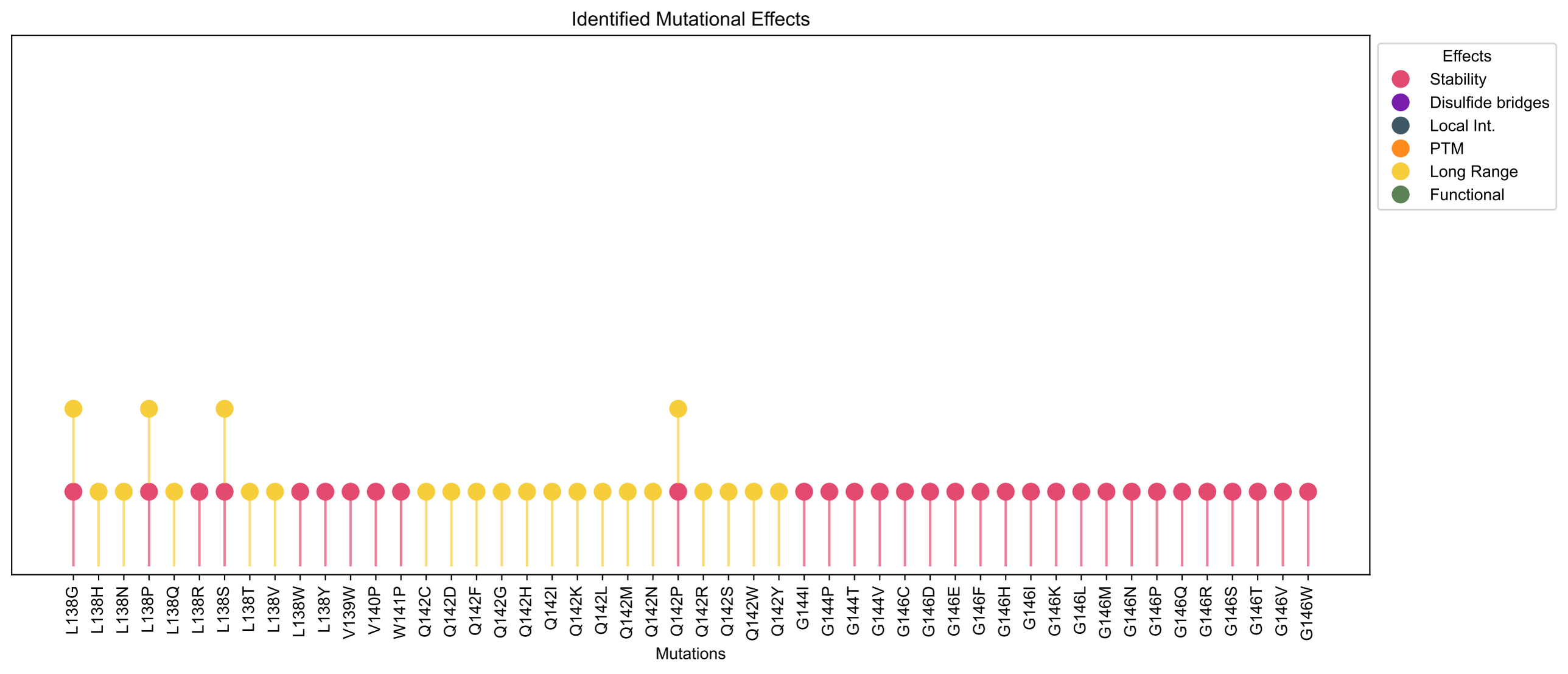

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
V92P
V92R
V92S
V92W
V92Y
I93D
I93G
I93P
I93S
K94P
A95D
A95E
A95F
A95H
A95I
A95K
A95L
A95M
A95N
A95P
A95Q
A95R
A95T
A95V
A95W
A95Y
F96A
F96C
F96D
F96E
F96G
F96K
F96N
F96P
F96Q
F96R
F96S
F96T
G97C
G97D
G97E
G97F
G97H
G97I
G97K
G97L
G97M
G97N
G97P
G97Q
Mutations

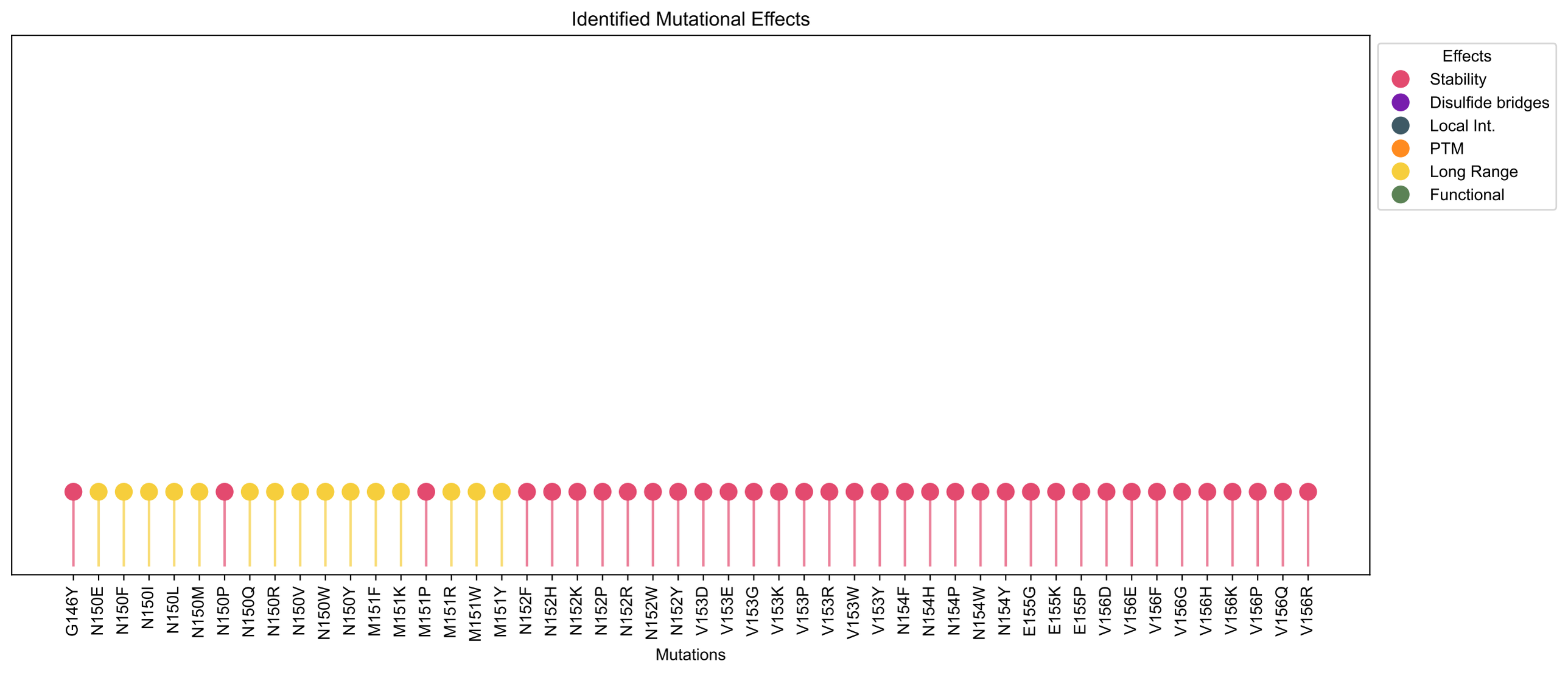

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
G97R
G97S
G97T
G97V
G97W
G97Y
I98D
I98G
I98P
L99C
L99D
L99E
L99G
L99K
L99N
L99P
L99R
L99S
K100P
R101D
R101P
A102D
A102E
A102F
A102H
A102I
A102K
A102L
A102M
A102N
A102P
A102Q
A102R
A102W
A102Y
A103P
A103W
A104D
A104E
A104F
A104H
A104K
A104P
A104R
A104W
A104Y
V106D
V106E
V106F
V106G
Mutations

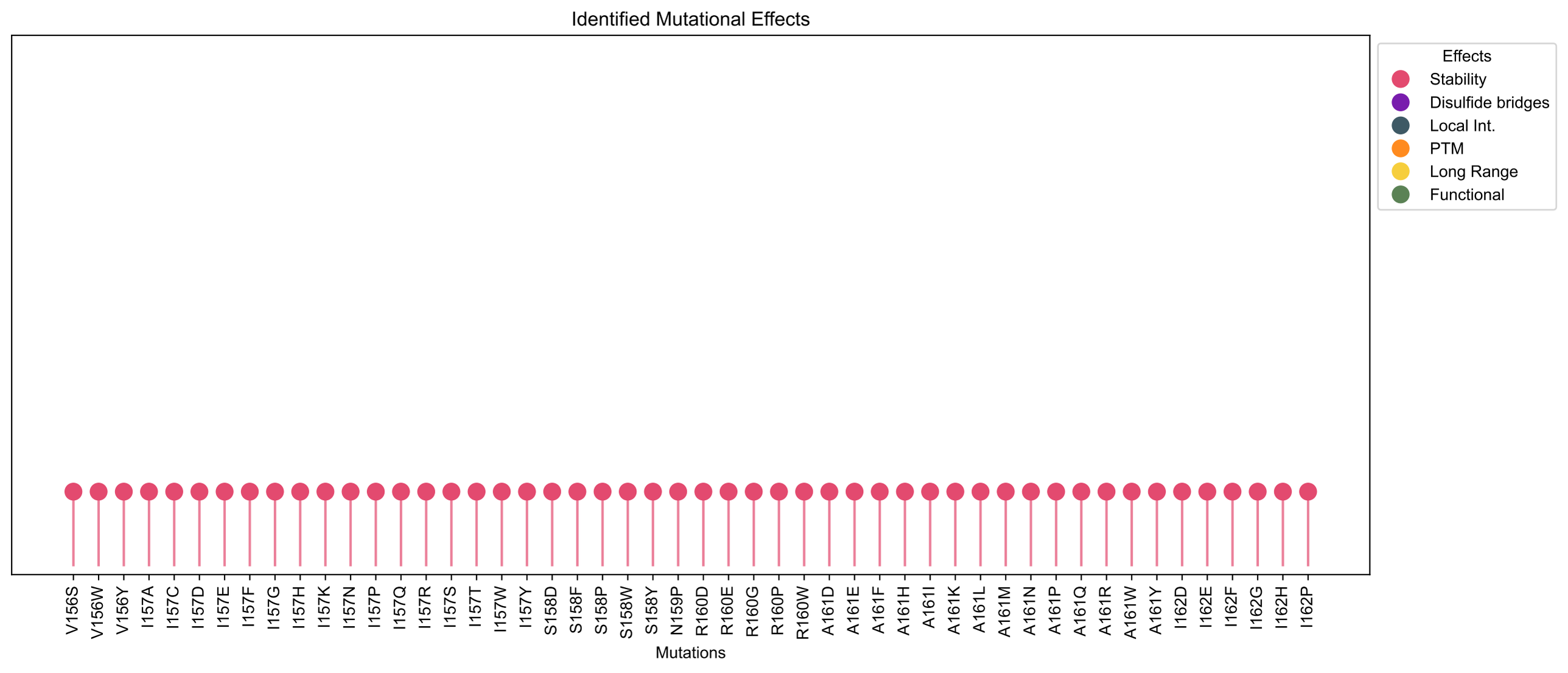

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
V106H
V106K
V106P
V106R
V106W
V106Y
N107P
Y110E
Y110K
Y110P
Y110R
G111P
L112D
L112G
L112P
L112S
L112W
L112Y
I116D
I116G
A117D
A117E
A117H
A117K
A117P
A117R
A117W
A117Y
N118P
A119D
A119P
I120A
I120D
I120E
I120F
I120G
I120H
I120K
I120N
I120P
I120R
I120S
I120W
I120Y
M121P
A123D
A123E
A123F
A123H
A123I
Mutations

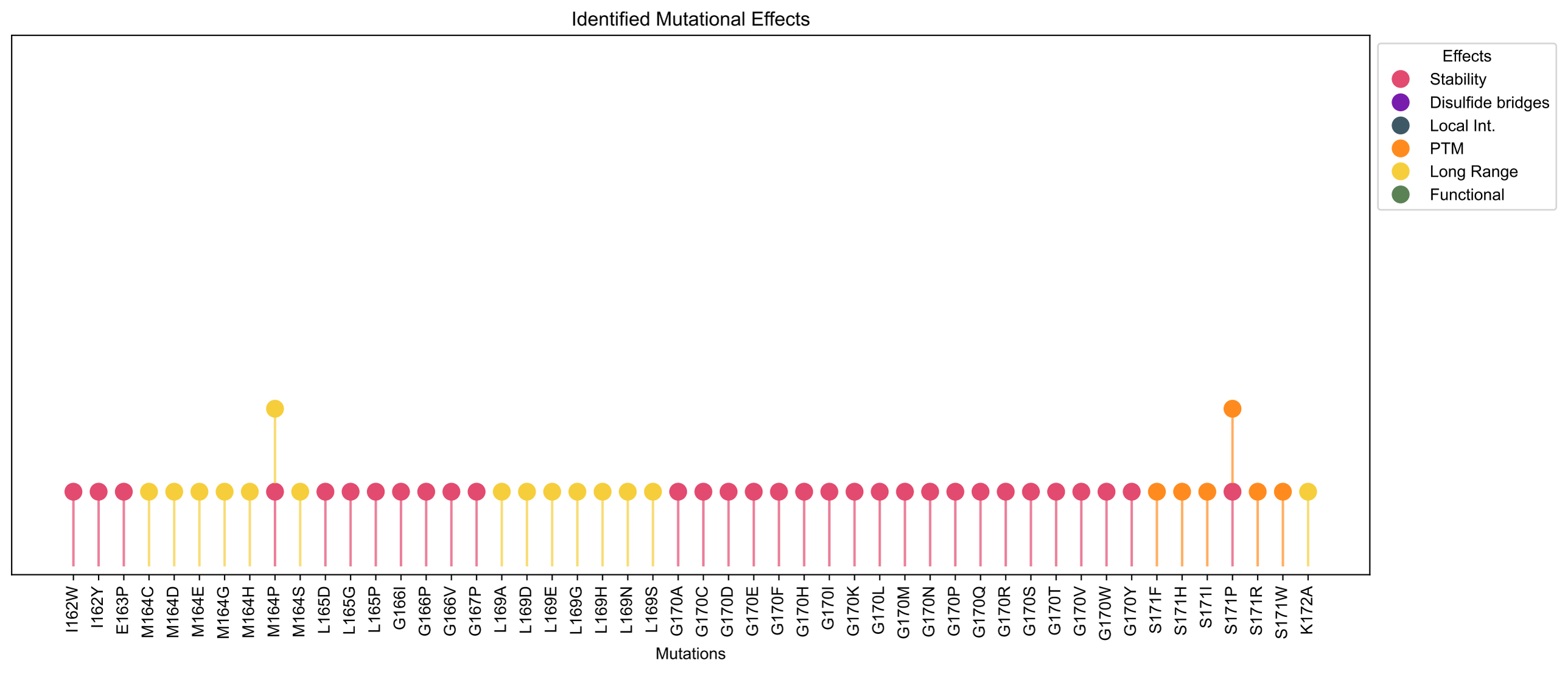

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
A123K
A123L
A123M
A123N
A123P
A123Q
A123R
A123W
A123Y
A124D
A124E
A124F
A124H
A124I
A124K
A124L
A124N
A124P
A124Q
A124R
A124W
A124Y
D125P
E126P
V127D
V127E
V127F
V127G
V127H
V127K
V127N
V127P
V127R
V127S
V127W
V127Y
A128D
A128P
E129P
G130I
G130P
G130T
G130V
K131P
L132A
L132D
L132E
L132G
L132K
L132N
Mutations

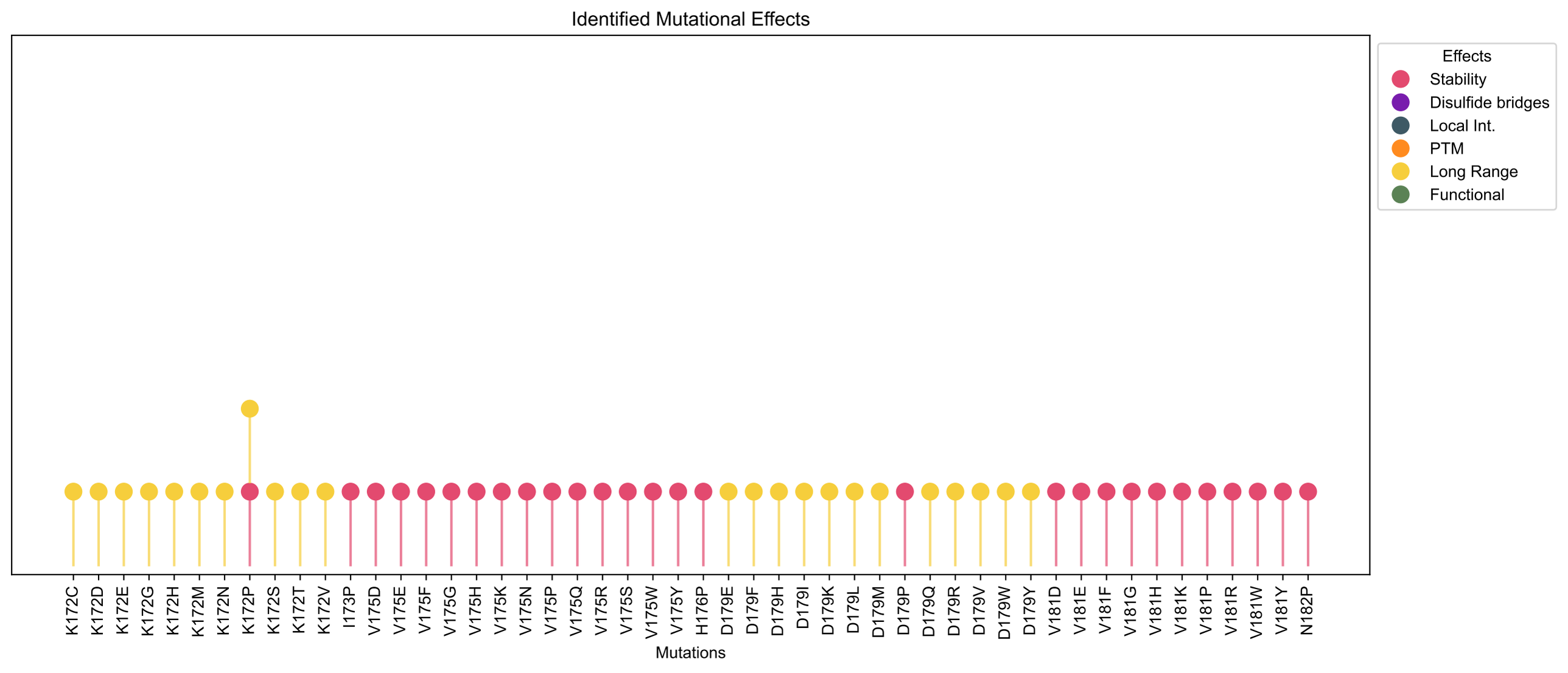

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
L132P
L132S
L132T
H135F
H135G
H135I
H135K
H135L
H135P
H135R
H135T
H135V
H135W
F136A
F136C
F136D
F136E
F136G
F136H
F136I
F136K
F136L
F136M
F136N
F136P
F136Q
F136R
F136S
F136T
F136V
F136W
P137A
P137E
P137F
P137G
P137H
P137I
P137K
P137L
P137M
P137Q
P137R
P137S
P137V
P137W
P137Y
L138A
L138C
L138D
L138E
Mutations

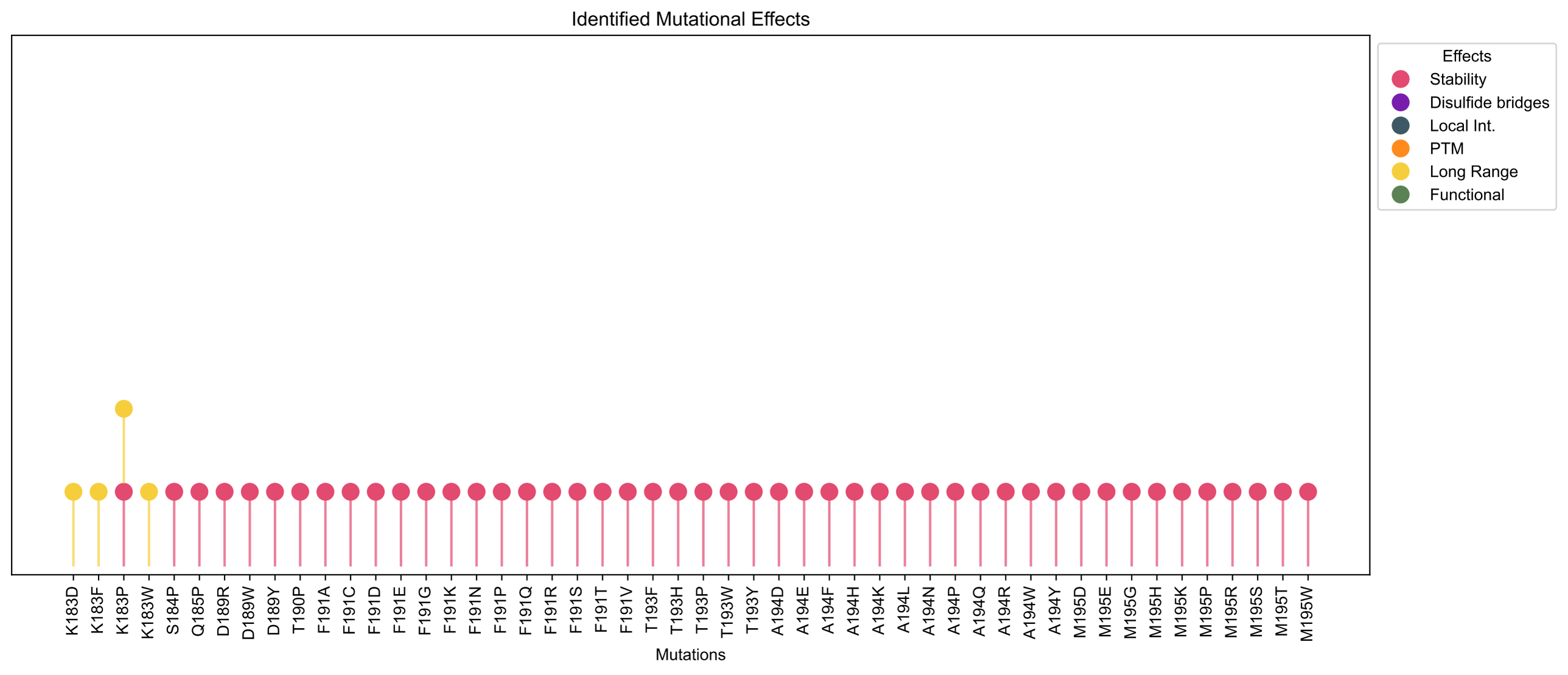

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
L138G
L138H
L138N
L138P
L138Q
L138R
L138S
L138T
L138V
L138W
L138Y
V139W
V140P
W141P
Q142C
Q142D
Q142F
Q142G
Q142H
Q142I
Q142K
Q142L
Q142M
Q142N
Q142P
Q142R
Q142S
Q142W
Q142Y
G144I
G144P
G144T
G144V
G146C
G146D
G146E
G146F
G146H
G146I
G146K
G146L
G146M
G146N
G146P
G146Q
G146R
G146S
G146T
G146V
G146W
Mutations

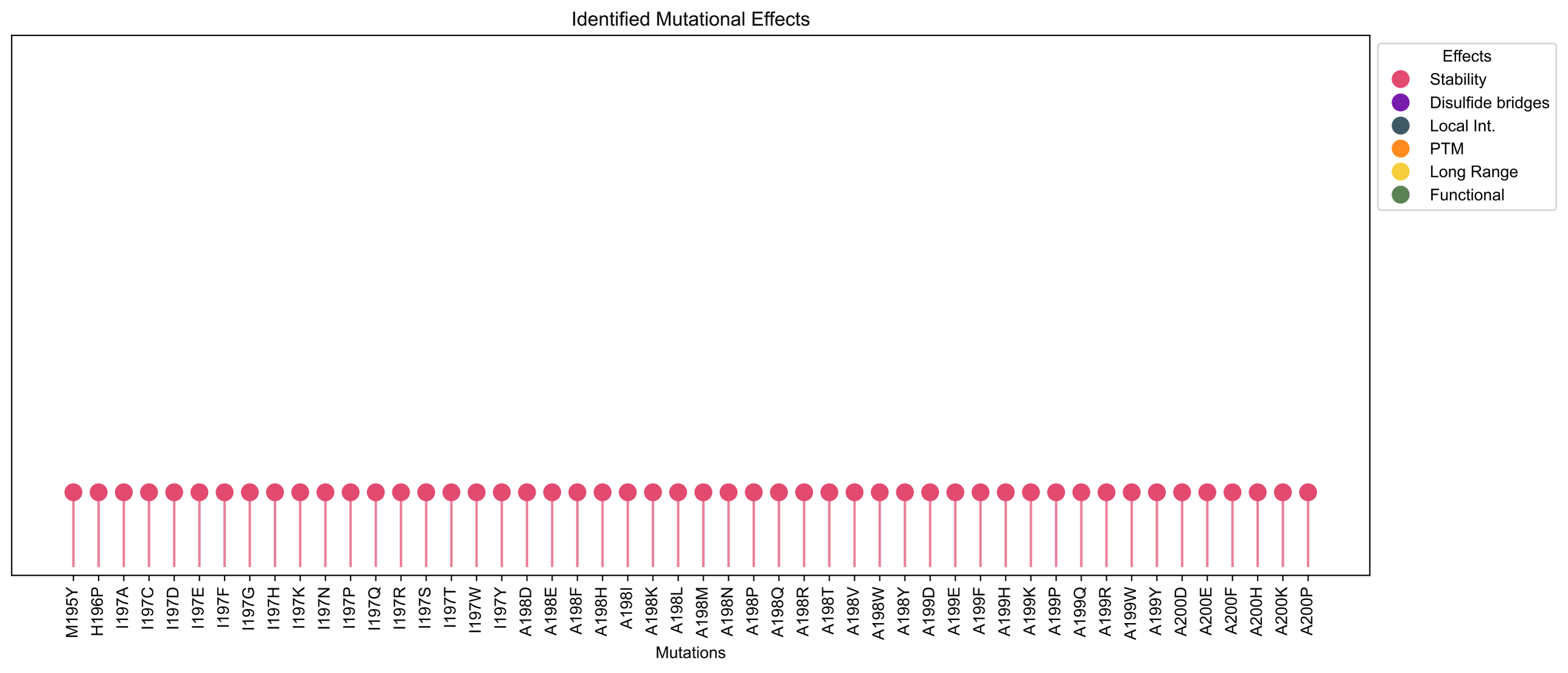

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
G146Y
N150E
N150F
N150I
N150L
N150M
N150P
N150Q
N150R
N150V
N150W
N150Y
M151F
M151K
M151P
M151R
M151W
M151Y
N152F
N152H
N152K
N152P
N152R
N152W
N152Y
V153D
V153E
V153G
V153K
V153P
V153R
V153W
V153Y
N154F
N154H
N154P
N154W
N154Y
E155G
E155K
E155P
V156D
V156E
V156F
V156G
V156H
V156K
V156P
V156Q
V156R
Mutations

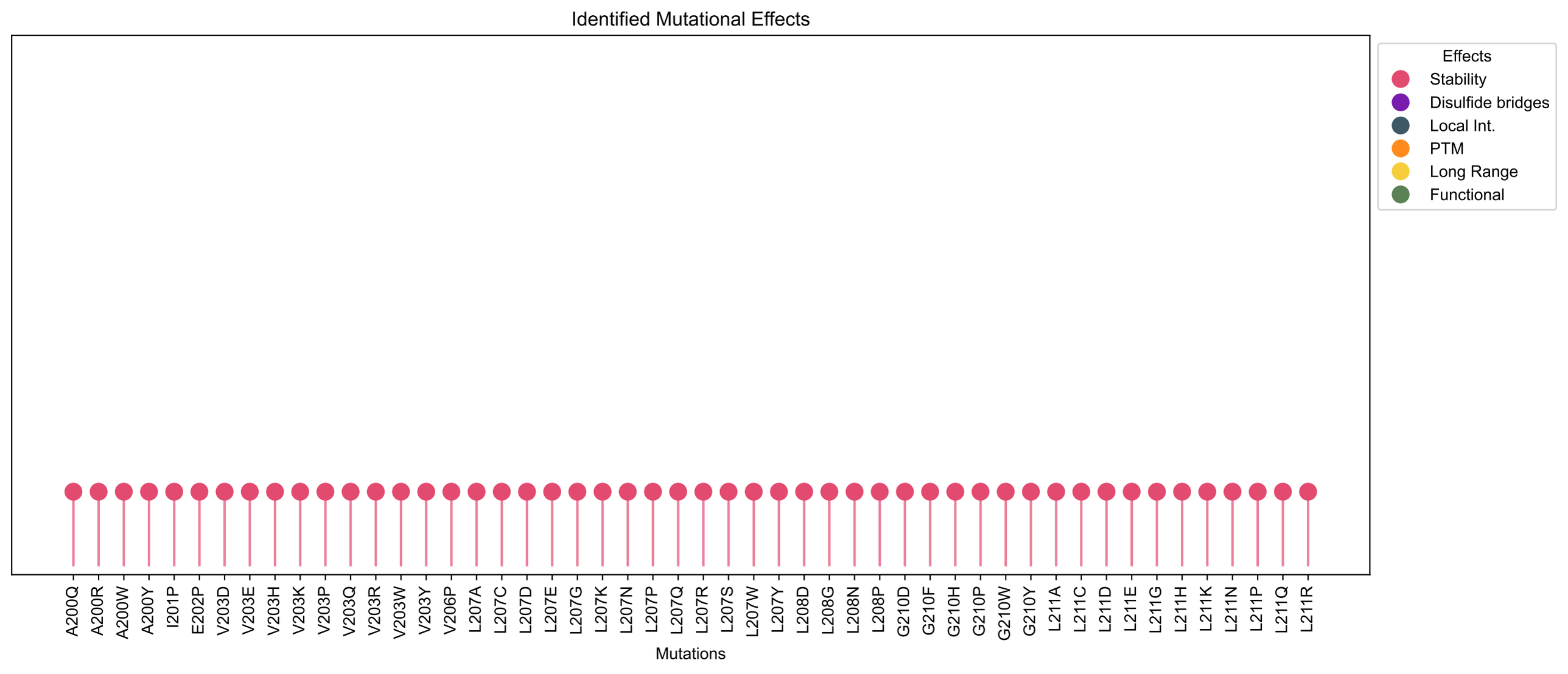

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
V156S
V156W
V156Y
I157A
I157C
I157D
I157E
I157F
I157G
I157H
I157K
I157N
I157P
I157Q
I157R
I157S
I157T
I157W
I157Y
S158D
S158F
S158P
S158W
S158Y
N159P
R160D
R160E
R160G
R160P
R160W
A161D
A161E
A161F
A161H
A161I
A161K
A161L
A161M
A161N
A161P
A161Q
A161R
A161W
A161Y
I162D
I162E
I162F
I162G
I162H
I162P
Mutations

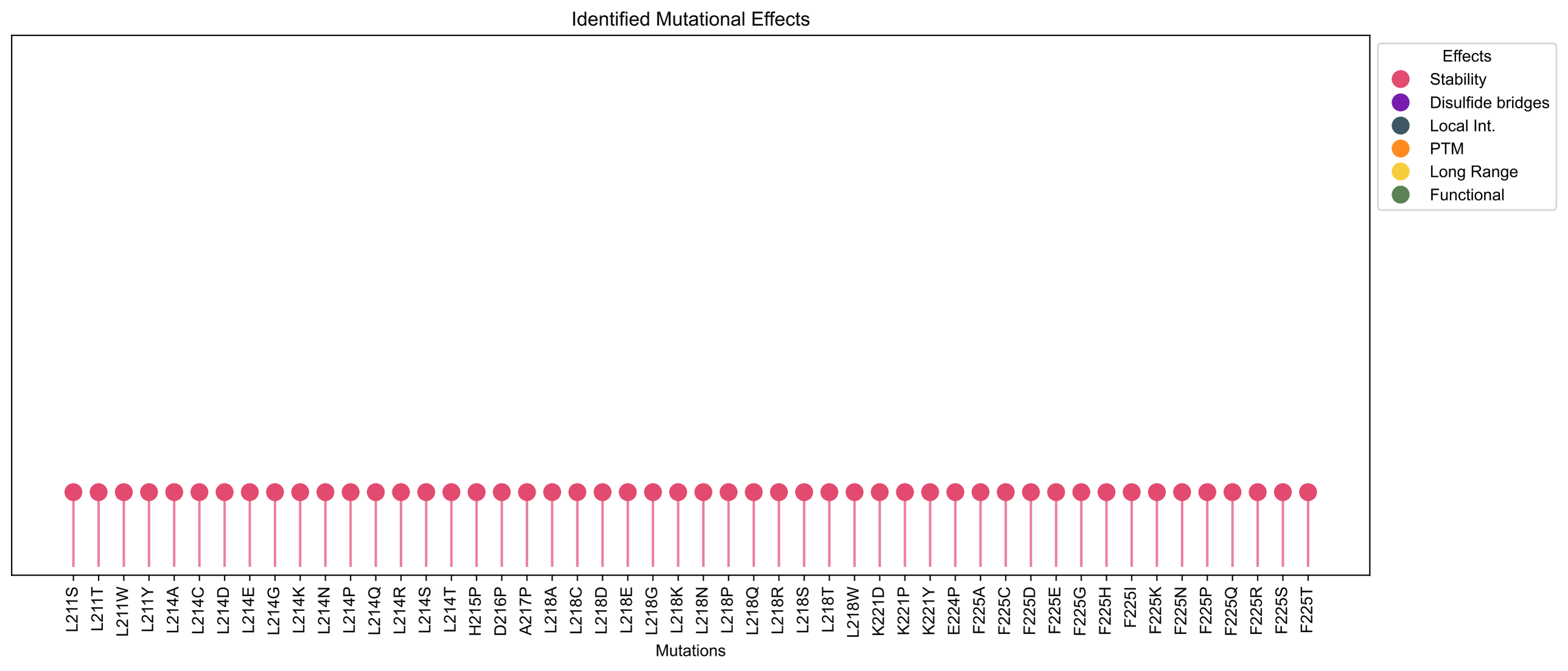

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
I162W
I162Y
E163P
M164C
M164D
M164E
M164G
M164H
M164P
M164S
L165D
L165G
L165P
G166I
G166P
G166V
G167P
L169A
L169D
L169E
L169G
L169H
L169N
L169S
G170A
G170C
G170D
G170E
G170F
G170H
G170I
G170K
G170L
G170M
G170N
G170P
G170Q
G170R
G170S
G170T
G170V
G170W
G170Y
S171F
S171H
S171I
S171P
S171R
S171W
K172A
Mutations

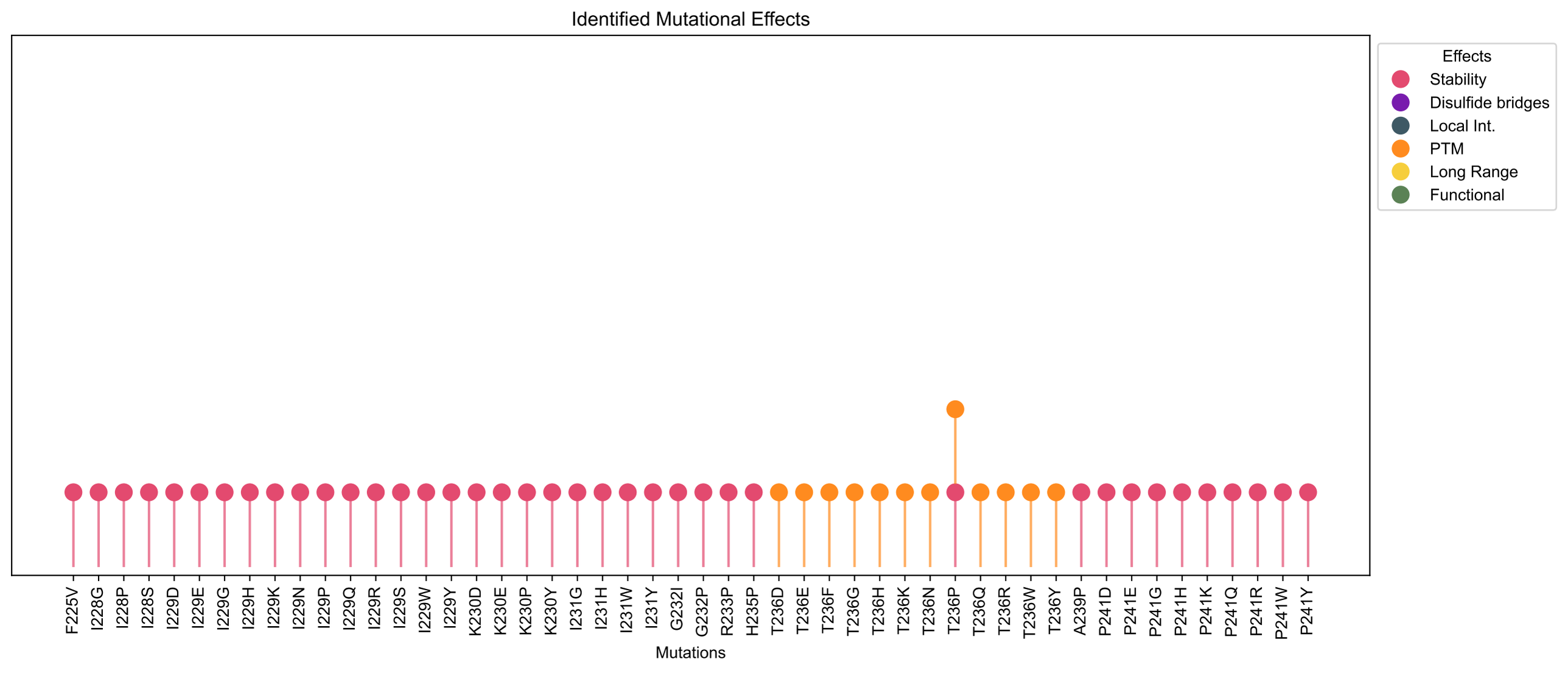

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
K172C
K172D
K172E
K172G
K172H
K172M
K172N
K172P
K172S
K172T
K172V
I173P
V175D
V175E
V175F
V175G
V175H
V175K
V175N
V175P
V175Q
V175R
V175S
V175W
V175Y
H176P
D179E
D179F
D179H
D179I
D179K
D179L
D179M
D179P
D179Q
D179R
D179V
D179W
D179Y
V181D
V181E
V181F
V181G
V181H
V181K
V181P
V181R
V181W
V181Y
N182P
Mutations

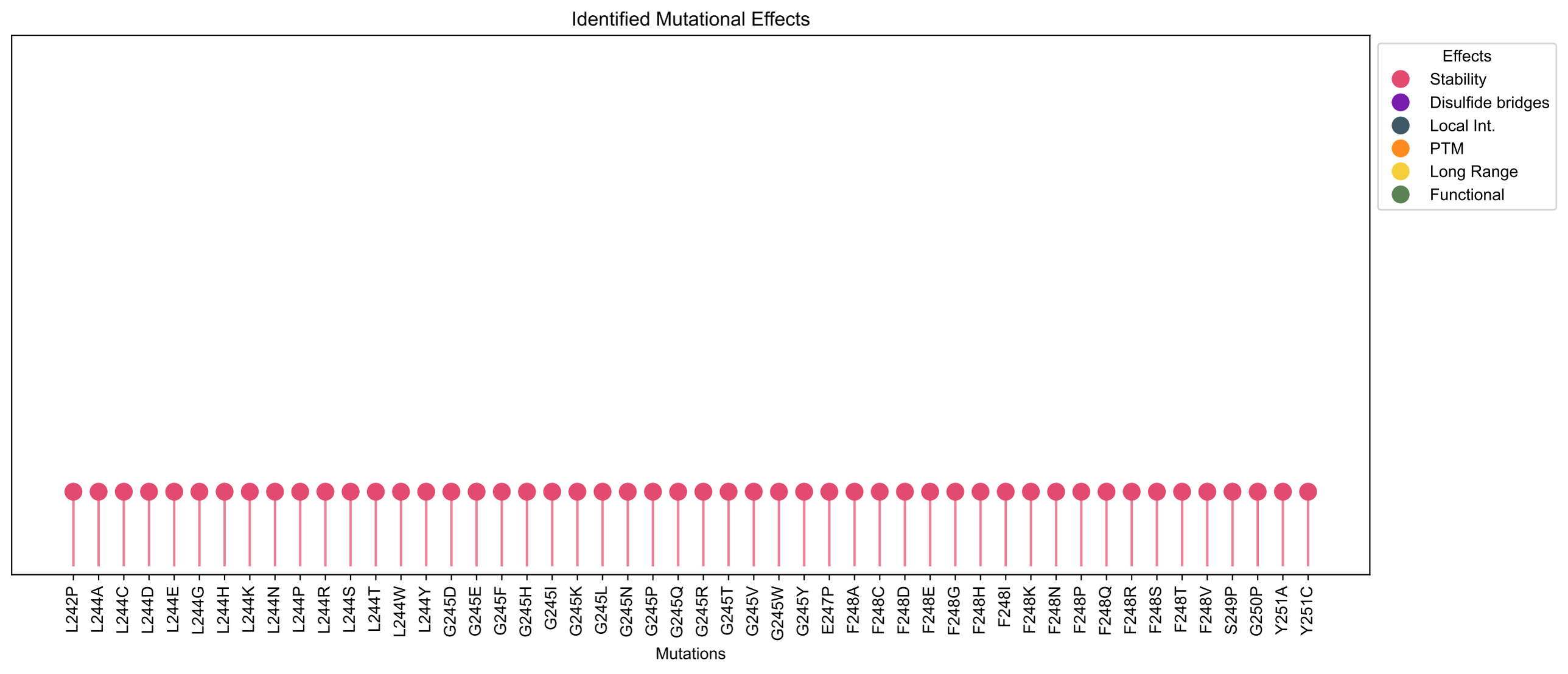

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
K183D
K183F
K183P
K183W
S184P
Q185P
D189R
D189W
D189Y
T190P
F191A
F191C
F191D
F191E
F191G
F191K
F191N
F191P
F191Q
F191R
F191S
F191T
F191V
T193F
T193H
T193P
T193W
T193Y
A194D
A194E
A194F
A194H
A194K
A194L
A194N
A194P
A194Q
A194R
A194W
A194Y
M195D
M195E
M195G
M195H
M195K
M195P
M195R
M195S
M195T
M195W
Mutations

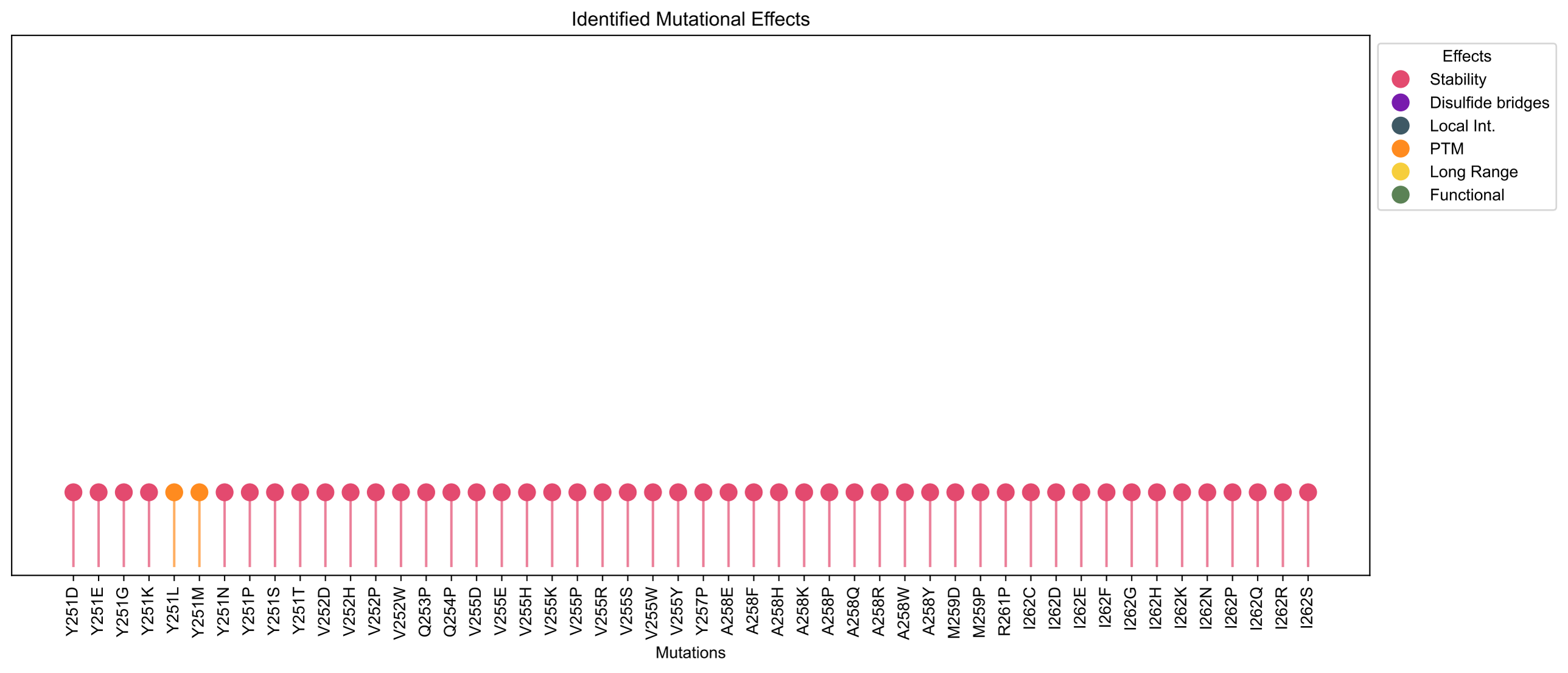

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
M195Y
H196P
I197A
I197C
I197D
I197E
I197F
I197G
I197H
I197K
I197N
I197P
I197Q
I197R
I197S
I197T
I197W
I197Y
A198D
A198E
A198F
A198H
A198I
A198K
A198L
A198M
A198N
A198P
A198Q
A198R
A198T
A198V
A198W
A198Y
A199D
A199E
A199F
A199H
A199K
A199P
A199Q
A199R
A199W
A199Y
A200D
A200E
A200F
A200H
A200K
A200P
Mutations

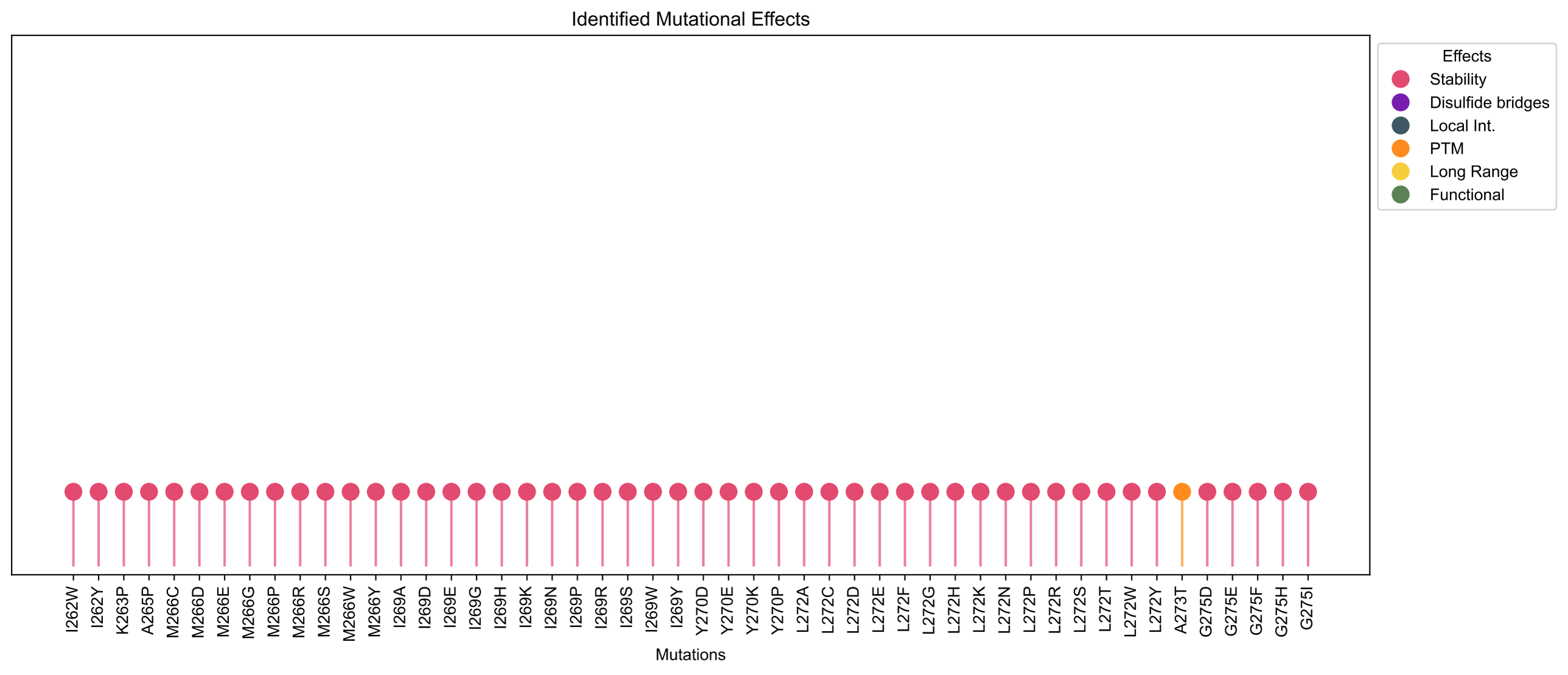

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
A200Q
A200R
A200W
A200Y
I201P
E202P
V203D
V203E
V203H
V203K
V203P
V203Q
V203R
V203W
V203Y
V206P
L207A
L207C
L207D
L207E
L207G
L207K
L207N
L207P
L207Q
L207R
L207S
L207W
L207Y
L208D
L208G
L208N
L208P
G210D
G210F
G210H
G210P
G210W
G210Y
L211A
L211C
L211D
L211E
L211G
L211H
L211K
L211N
L211P
L211Q
L211R
Mutations

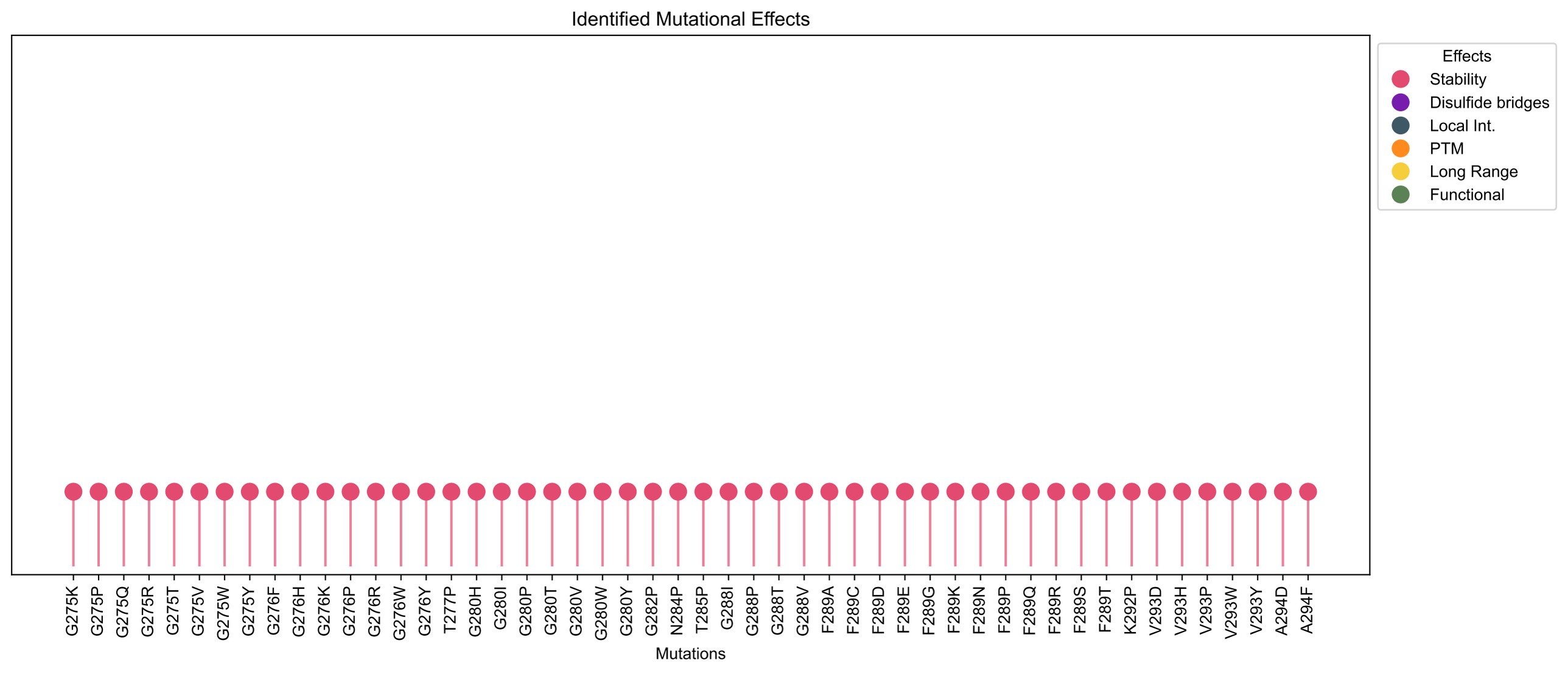

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
L211S
L211T
L211W
L211Y
L214A
L214C
L214D
L214E
L214G
L214K
L214N
L214P
L214Q
L214R
L214S
L214T
H215P
D216P
A217P
L218A
L218C
L218D
L218E
L218G
L218K
L218N
L218P
L218Q
L218R
L218S
L218T
L218W
K221D
K221P
K221Y
E224P
F225A
F225C
F225D
F225E
F225G
F225H
F225I
F225K
F225N
F225P
F225Q
F225R
F225S
F225T
Mutations

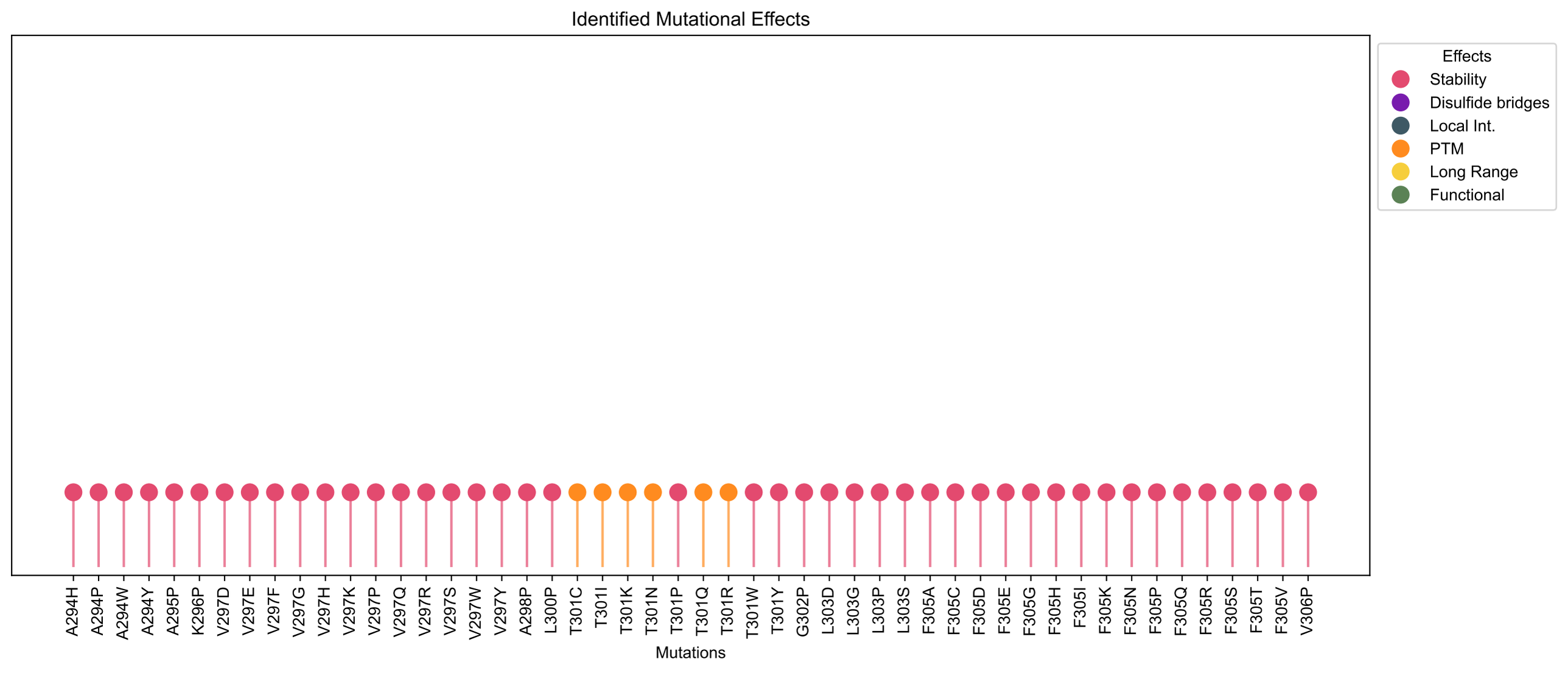

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
F225V
I228G
I228P
I228S
I229D
I229E
I229G
I229H
I229K
I229N
I229P
I229Q
I229R
I229S
I229W
I229Y
K230D
K230E
K230P
K230Y
I231G
I231H
I231W
I231Y
G232I
G232P
R233P
H235P
T236D
T236E
T236F
T236G
T236H
T236K
T236N
T236P
T236Q
T236R
T236W
T236Y
A239P
P241D
P241E
P241G
P241H
P241K
P241Q
P241R
P241W
P241Y
Mutations

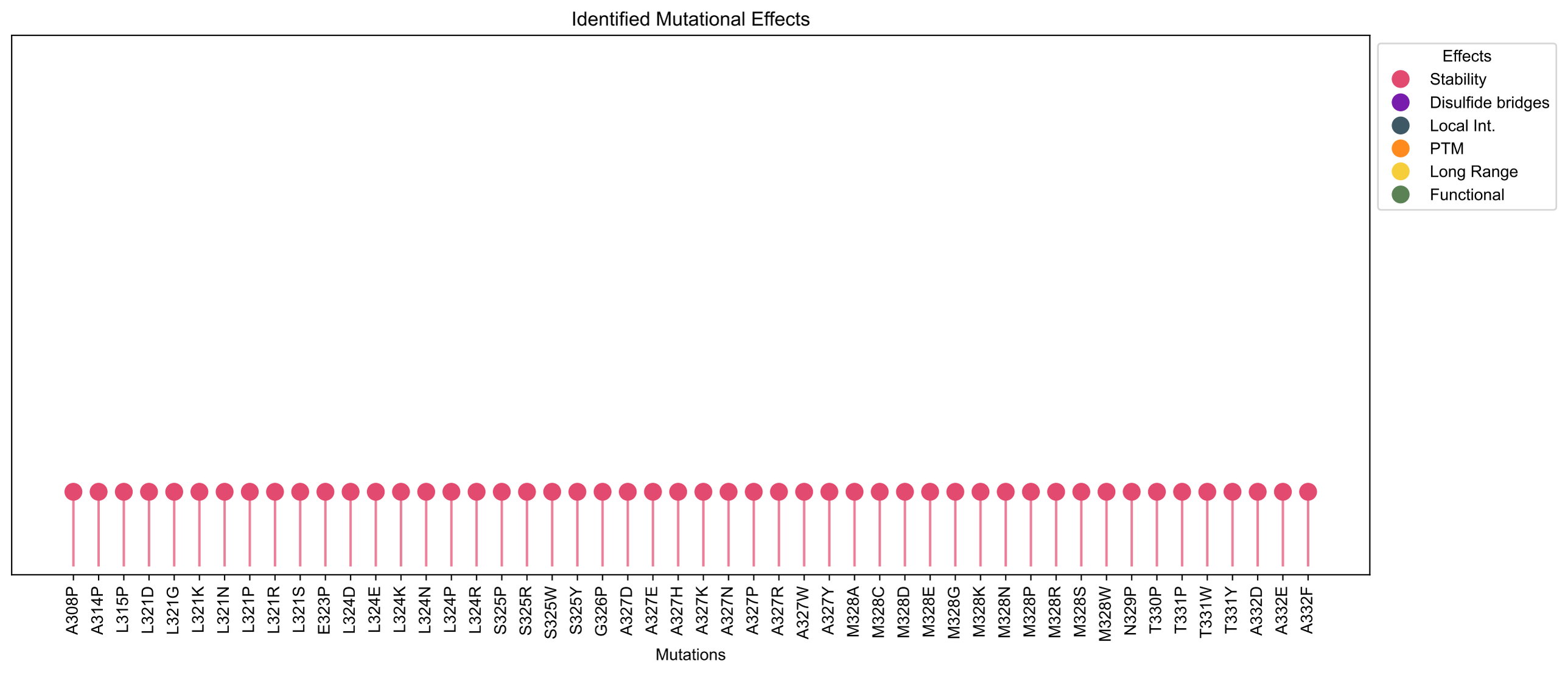

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
L242P
L244A
L244C
L244D
L244E
L244G
L244H
L244K
L244N
L244P
L244R
L244S
L244T
L244W
L244Y
G245D
G245E
G245F
G245H
G245I
G245K
G245L
G245N
G245P
G245Q
G245R
G245T
G245V
G245W
G245Y
E247P
F248A
F248C
F248D
F248E
F248G
F248H
F248I
F248K
F248N
F248P
F248Q
F248R
F248S
F248T
F248V
S249P
G250P
Y251A
Y251C
Mutations

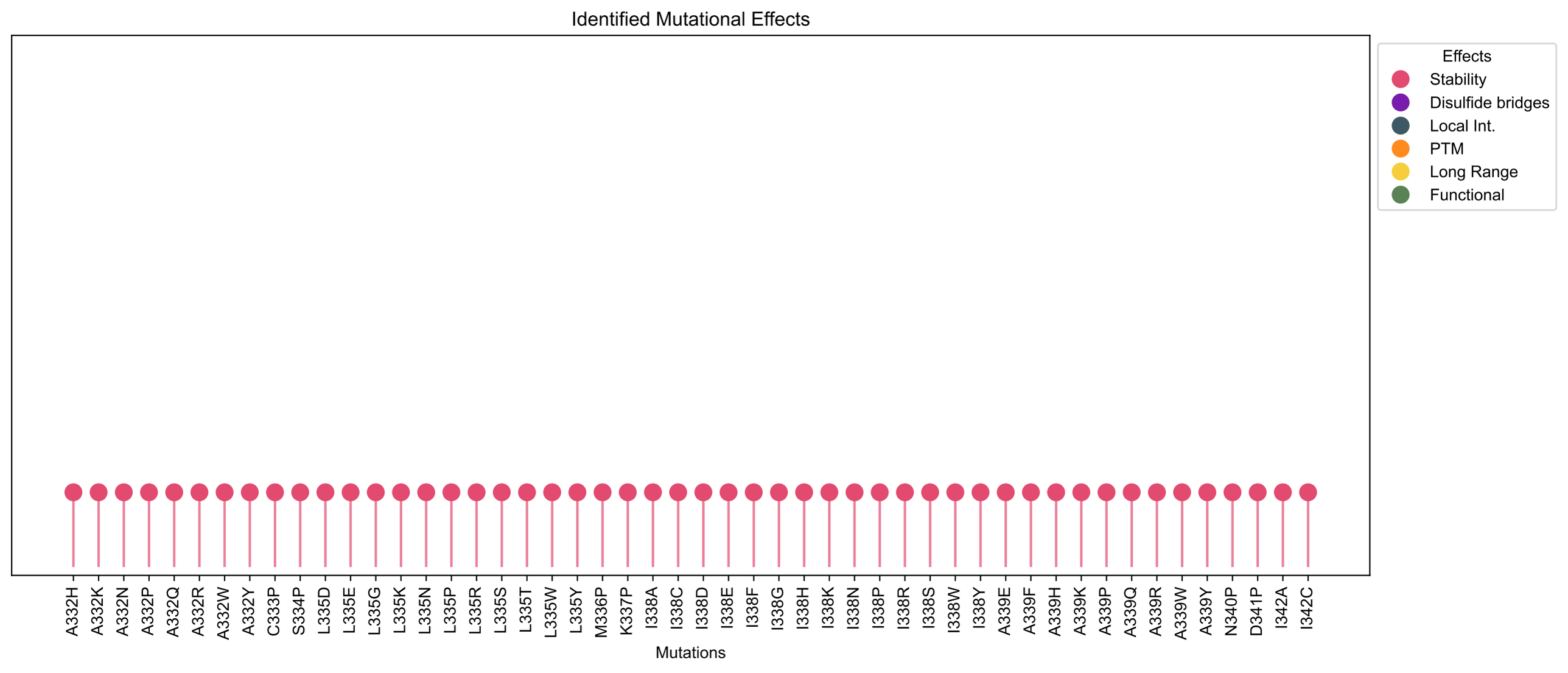

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
Y251D
Y251E
Y251G
Y251K
Y251L
Y251M
Y251N
Y251P
Y251S
Y251T
V252D
V252H
V252P
V252W
Q253P
Q254P
V255D
V255E
V255H
V255K
V255P
V255R
V255S
V255W
V255Y
Y257P
A258E
A258F
A258H
A258K
A258P
A258Q
A258R
A258W
A258Y
M259D
M259P
R261P
I262C
I262D
I262E
I262F
I262G
I262H
I262K
I262N
I262P
I262Q
I262R
I262S
Mutations

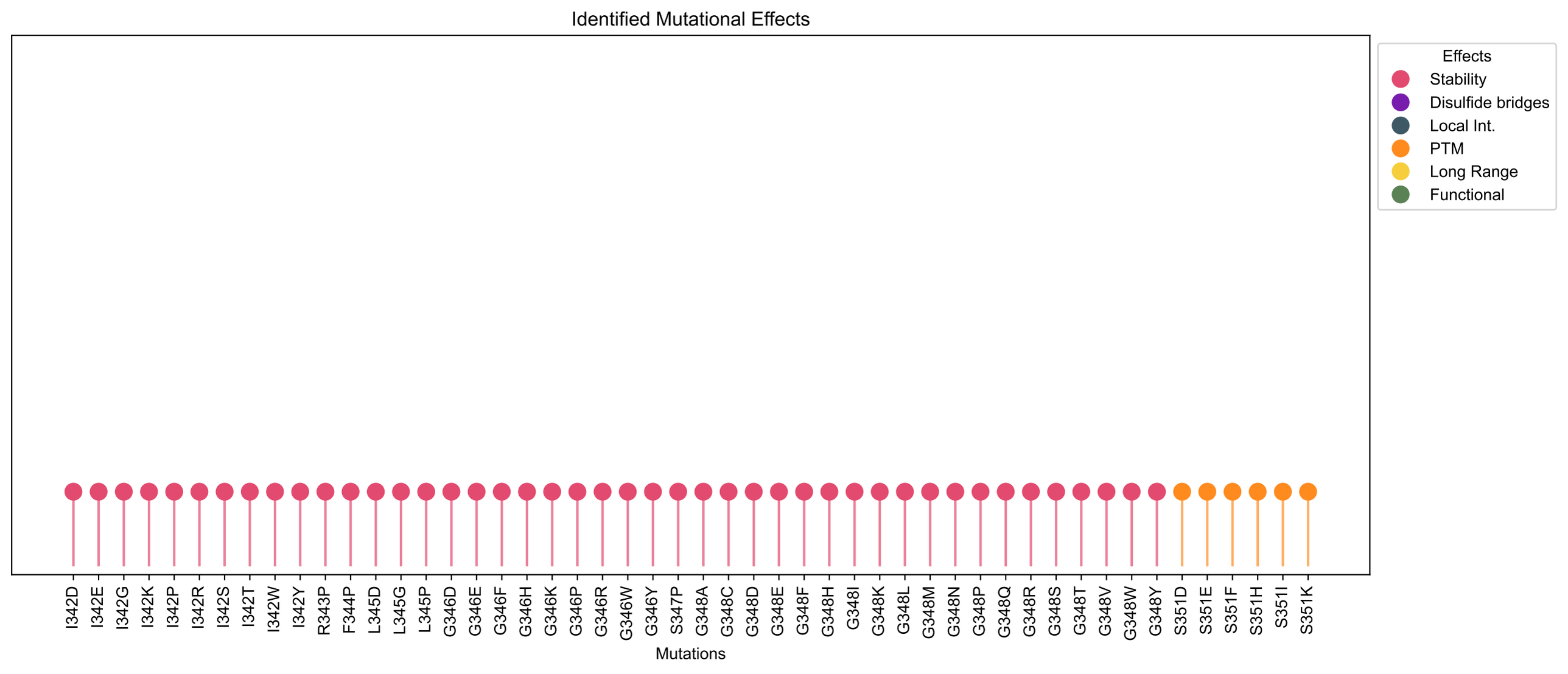

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
I262W
I262Y
K263P
A265P
M266C
M266D
M266E
M266G
M266P
M266R
M266S
M266W
M266Y
I269A
I269D
I269E
I269G
I269H
I269K
I269N
I269P
I269R
I269S
I269W
I269Y
Y270D
Y270E
Y270K
Y270P
L272A
L272C
L272D
L272E
L272F
L272G
L272H
L272K
L272N
L272P
L272R
L272S
L272T
L272W
L272Y
A273T
G275D
G275E
G275F
G275H
G275I
Mutations

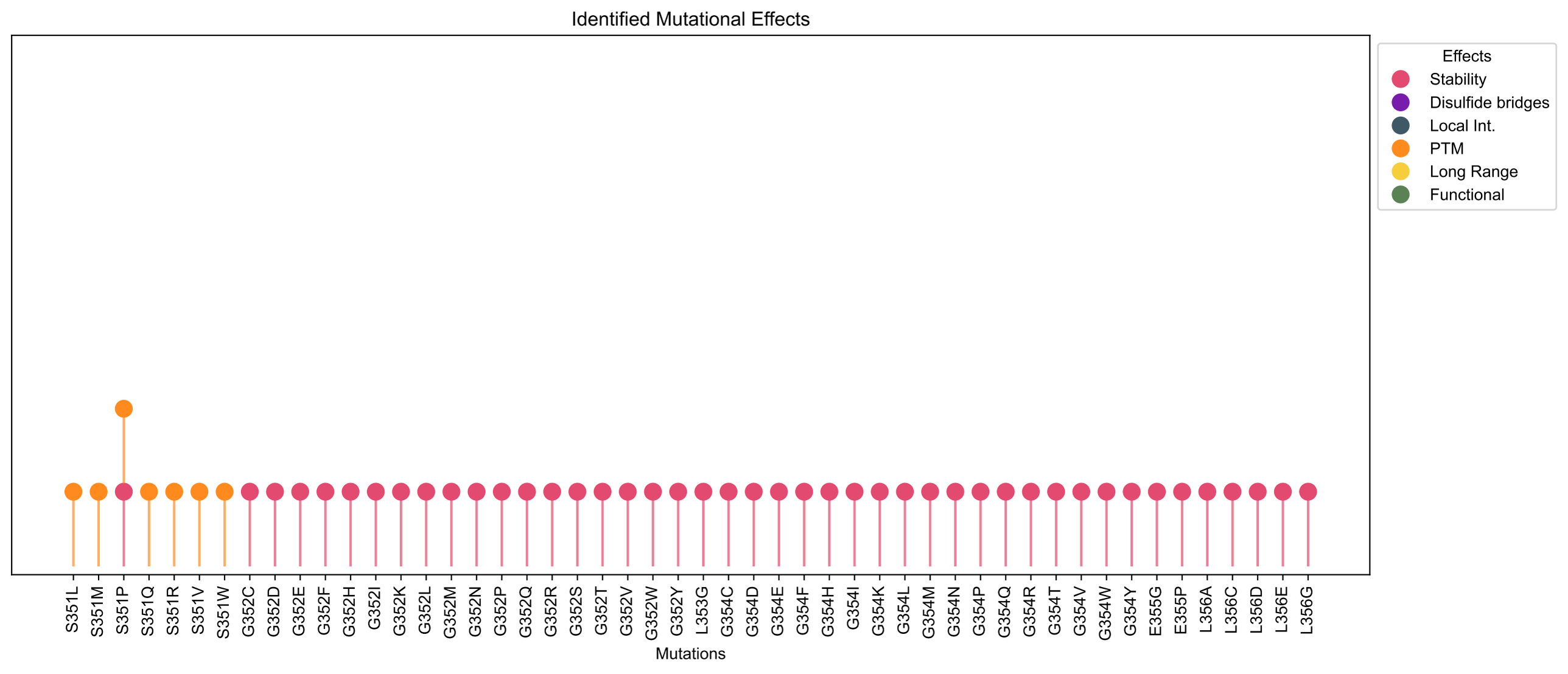

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
G275K
G275P
G275Q
G275R
G275T
G275V
G275W
G275Y
G276F
G276H
G276K
G276P
G276R
G276W
G276Y
T277P
G280H
G280I
G280P
G280T
G280V
G280W
G280Y
G282P
N284P
T285P
G288I
G288P
G288T
G288V
F289A
F289C
F289D
F289E
F289G
F289K
F289N
F289P
F289Q
F289R
F289S
F289T
K292P
V293D
V293H
V293P
V293W
V293Y
A294D
A294F
Mutations

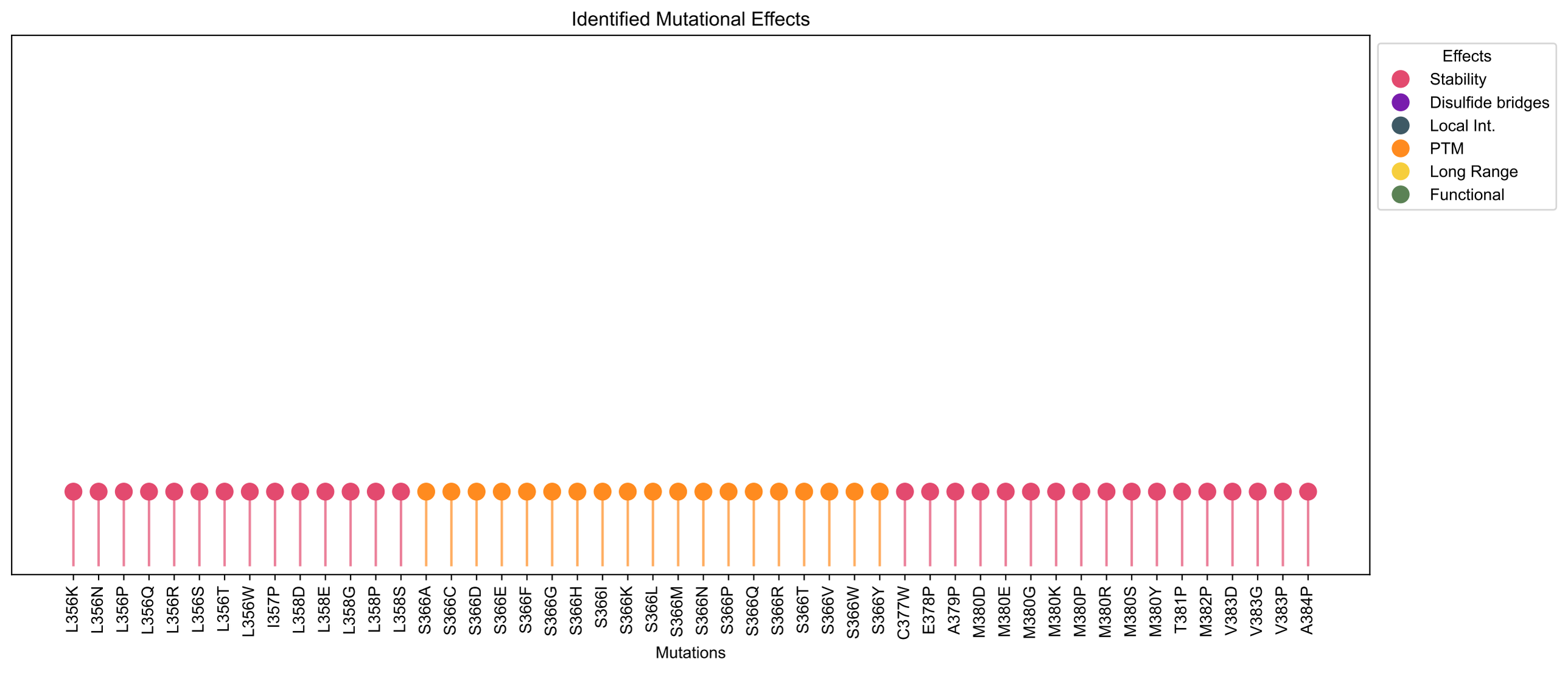

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
A294H
A294P
A294W
A294Y
A295P
K296P
V297D
V297E
V297F
V297G
V297H
V297K
V297P
V297Q
V297R
V297S
V297W
V297Y
A298P
L300P
T301C
T301I
T301K
T301N
T301P
T301Q
T301R
T301W
T301Y
G302P
L303D
L303G
L303P
L303S
F305A
F305C
F305D
F305E
F305G
F305H
F305I
F305K
F305N
F305P
F305Q
F305R
F305S
F305T
F305V
V306P
Mutations

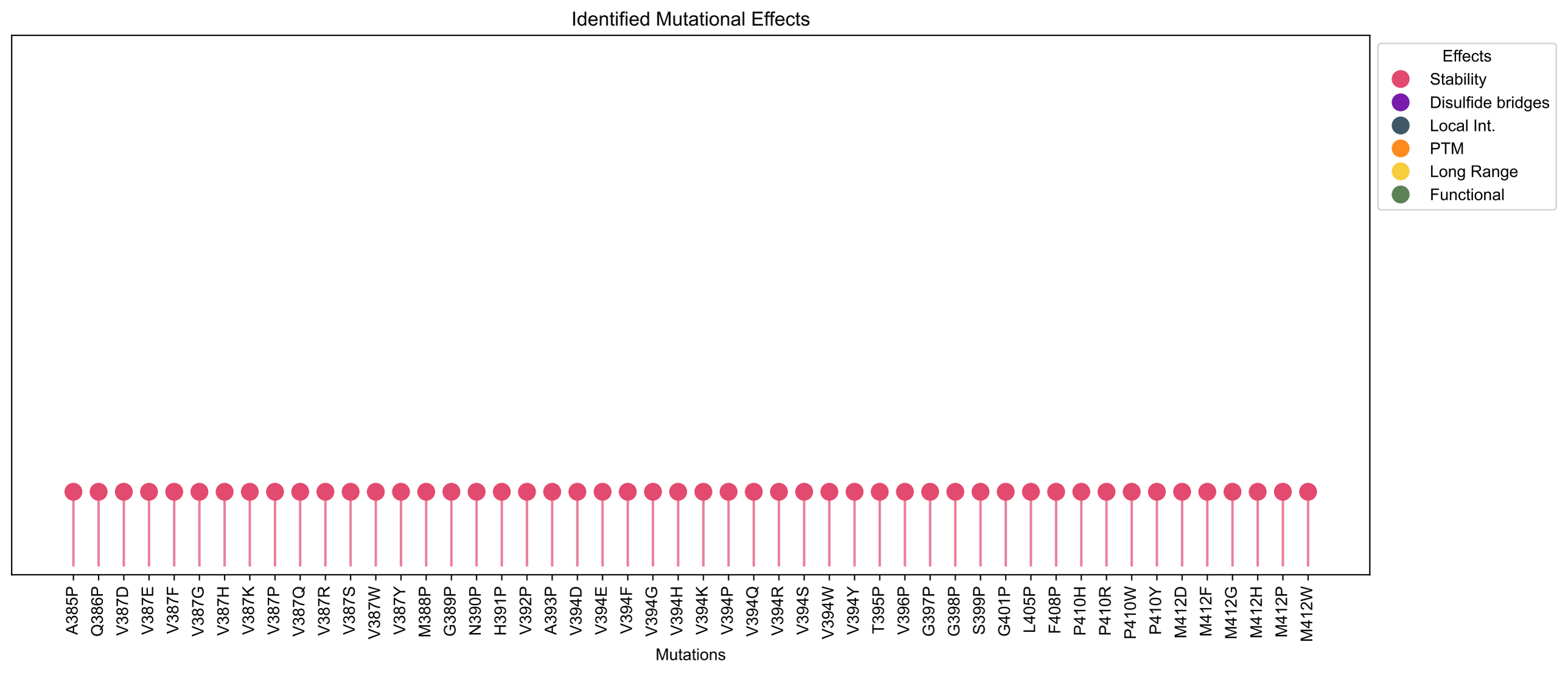

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
A308P
A314P
L315P
L321D
L321G
L321K
L321N
L321P
L321R
L321S
E323P
L324D
L324E
L324K
L324N
L324P
L324R
S325P
S325R
S325W
S325Y
G326P
A327D
A327E
A327H
A327K
A327N
A327P
A327R
A327W
A327Y
M328A
M328C
M328D
M328E
M328G
M328K
M328N
M328P
M328R
M328S
M328W
N329P
T330P
T331P
T331W
T331Y
A332D
A332E
A332F
Mutations

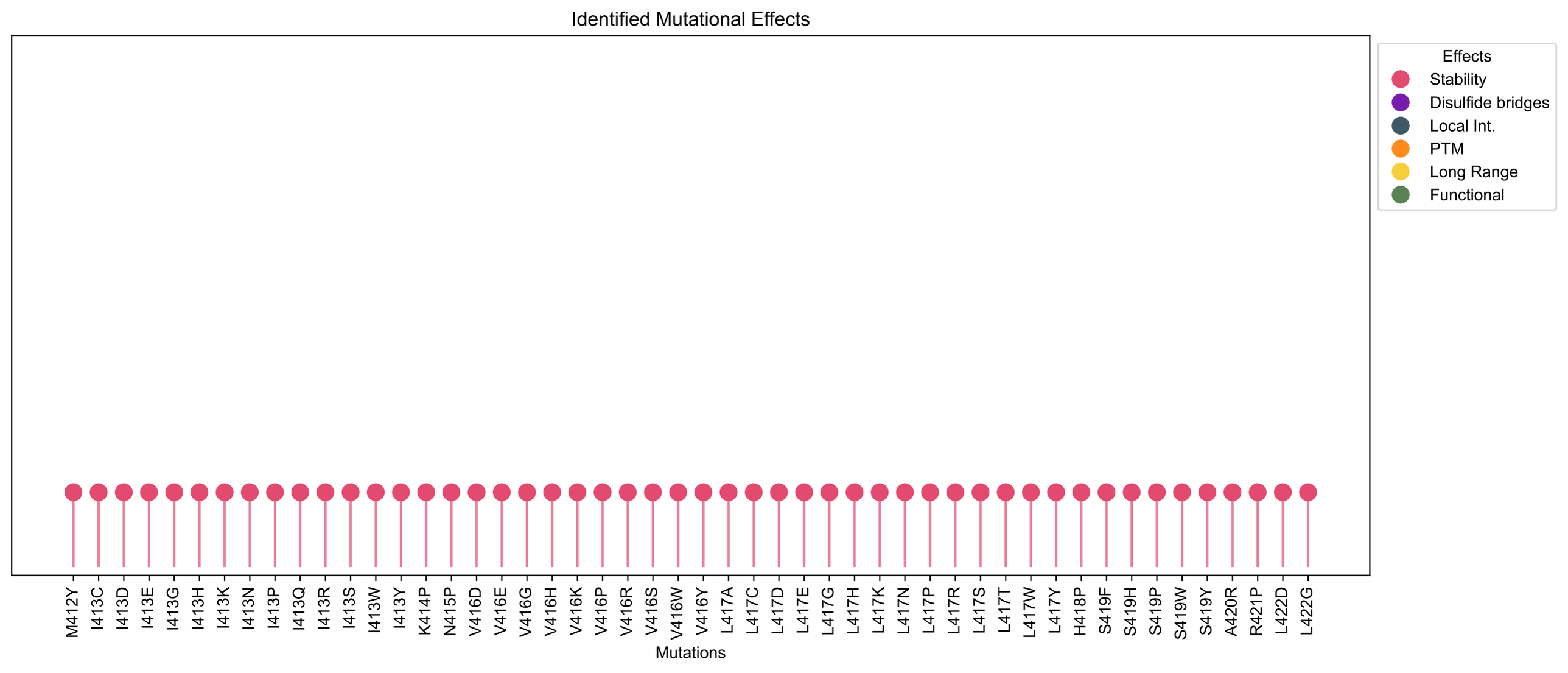

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
A332H
A332K
A332N
A332P
A332Q
A332R
A332W
A332Y
C333P
S334P
L335D
L335E
L335G
L335K
L335N
L335P
L335R
L335S
L335T
L335W
L335Y
M336P
K337P
I338A
I338C
I338D
I338E
I338F
I338G
I338H
I338K
I338N
I338P
I338R
I338S
I338W
I338Y
A339E
A339F
A339H
A339K
A339P
A339Q
A339R
A339W
A339Y
N340P
D341P
I342A
I342C
Mutations

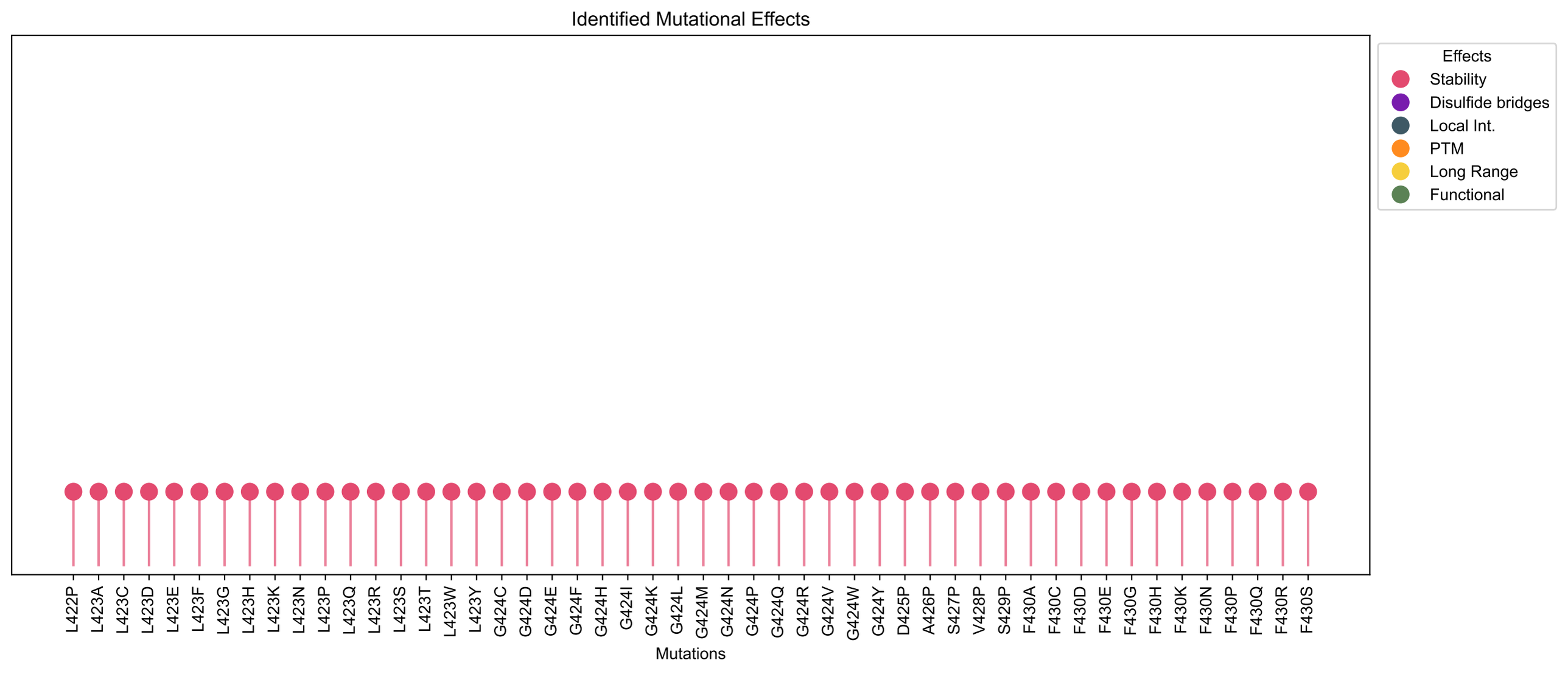

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
I342D
I342E
I342G
I342K
I342P
I342R
I342S
I342T
I342W
I342Y
R343P
F344P
L345D
L345G
L345P
G346D
G346E
G346F
G346H
G346K
G346P
G346R
G346W
G346Y
S347P
G348A
G348C
G348D
G348E
G348F
G348H
G348I
G348K
G348L
G348M
G348N
G348P
G348Q
G348R
G348S
G348T
G348V
G348W
G348Y
S351D
S351E
S351F
S351H
S351I
S351K
Mutations

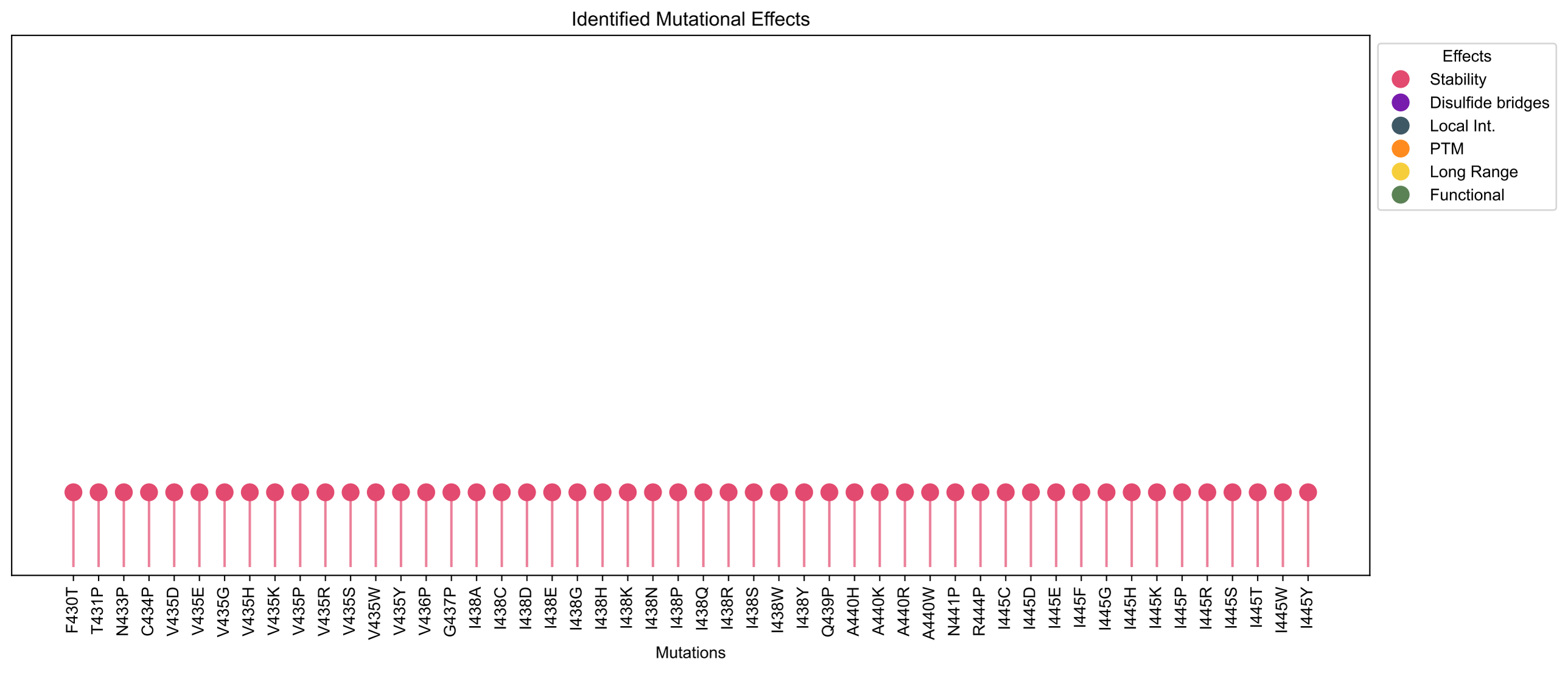

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
S351L
S351M
S351P
S351Q
S351R
S351V
S351W
G352C
G352D
G352E
G352F
G352H
G352I
G352K
G352L
G352M
G352N
G352P
G352Q
G352R
G352S
G352T
G352V
G352W
G352Y
L353G
G354C
G354D
G354E
G354F
G354H
G354I
G354K
G354L
G354M
G354N
G354P
G354Q
G354R
G354T
G354V
G354W
G354Y
E355G
E355P
L356A
L356C
L356D
L356E
L356G
Mutations

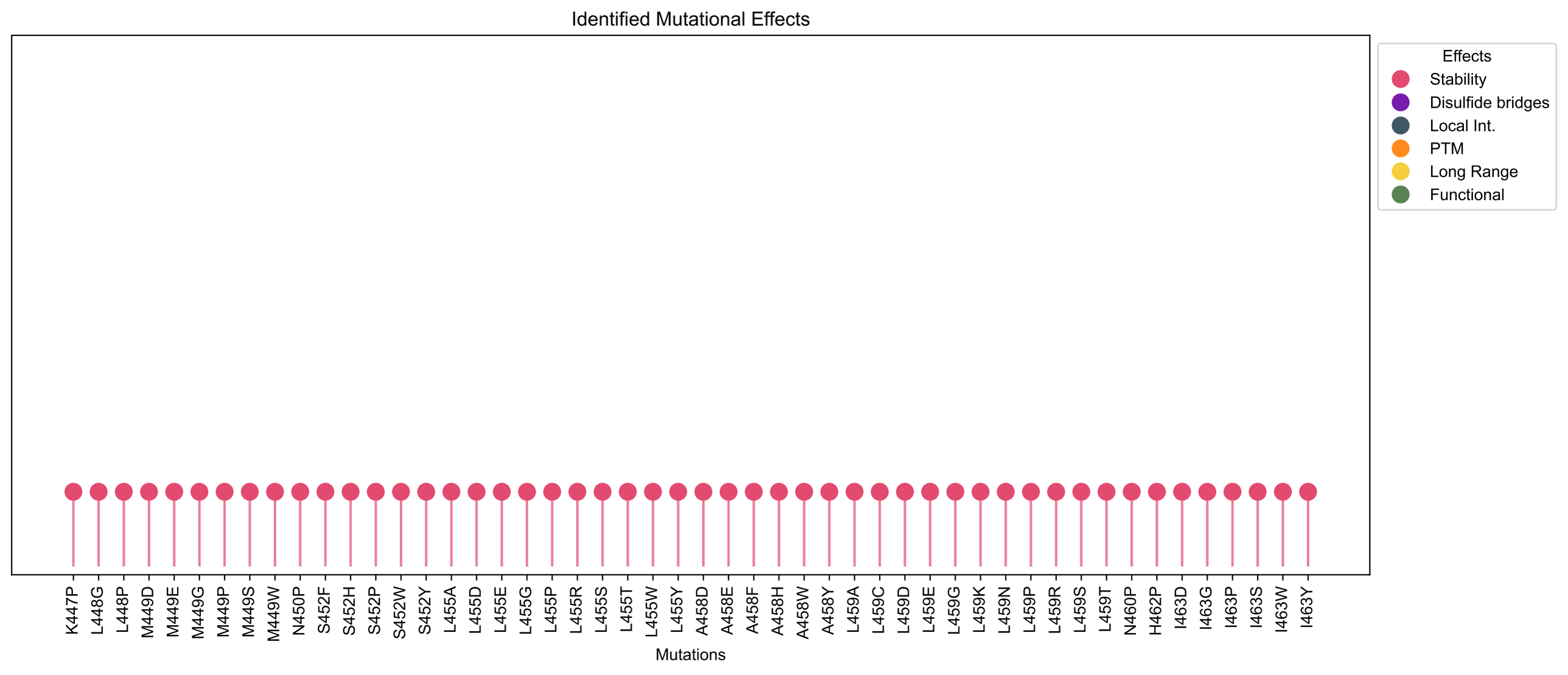

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
L356K
L356N
L356P
L356Q
L356R
L356S
L356T
L356W
I357P
L358D
L358E
L358G
L358P
L358S
S366A
S366C
S366D
S366E
S366F
S366G
S366H
S366I
S366K
S366L
S366M
S366N
S366P
S366Q
S366R
S366T
S366V
S366W
S366Y
C377W
E378P
A379P
M380D
M380E
M380G
M380K
M380P
M380R
M380S
M380Y
T381P
M382P
V383D
V383G
V383P
A384P
Mutations

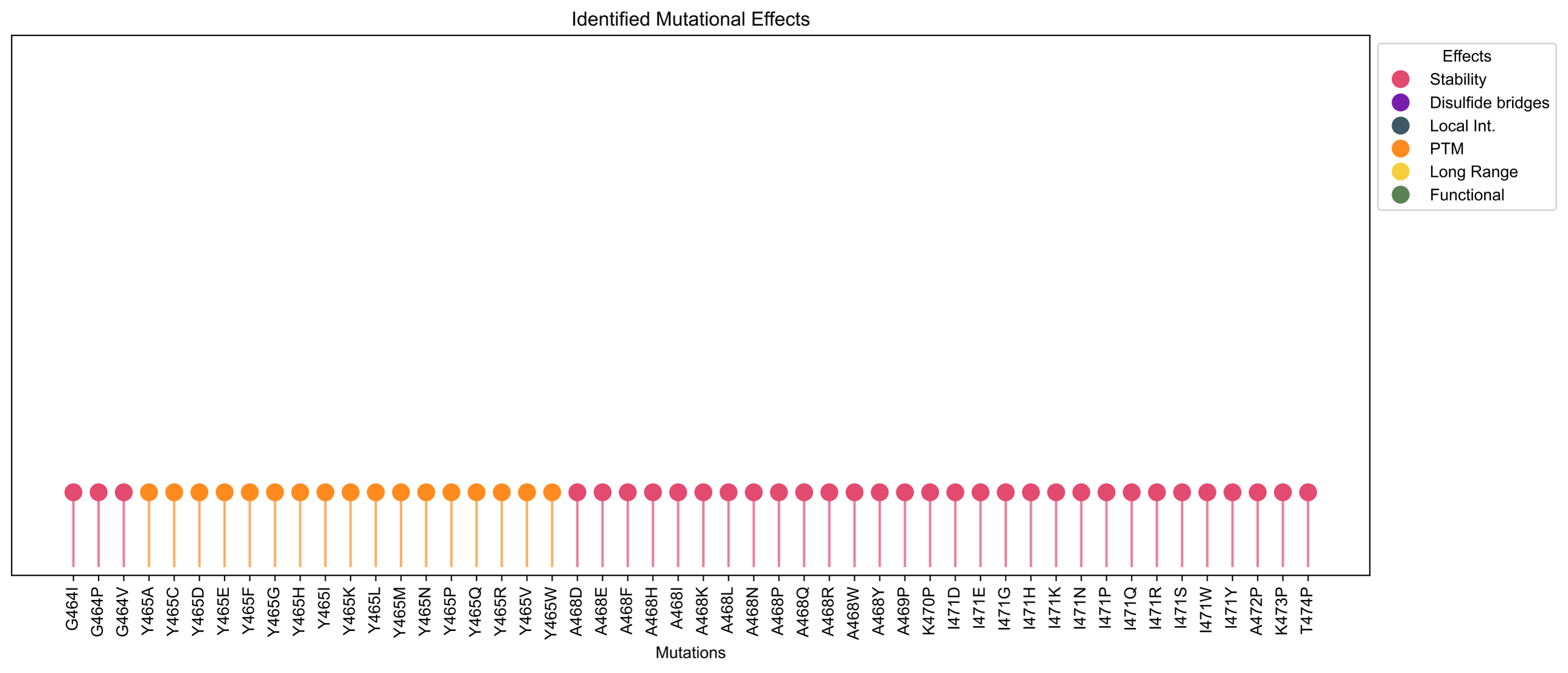

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
A385P
Q386P
V387D
V387E
V387F
V387G
V387H
V387K
V387P
V387Q
V387R
V387S
V387W
V387Y
M388P
G389P
N390P
H391P
V392P
A393P
V394D
V394E
V394F
V394G
V394H
V394K
V394P
V394Q
V394R
V394S
V394W
V394Y
T395P
V396P
G397P
G398P
S399P
G401P
L405P
F408P
P410H
P410R
P410W
P410Y
M412D
M412F
M412G
M412H
M412P
M412W
Mutations

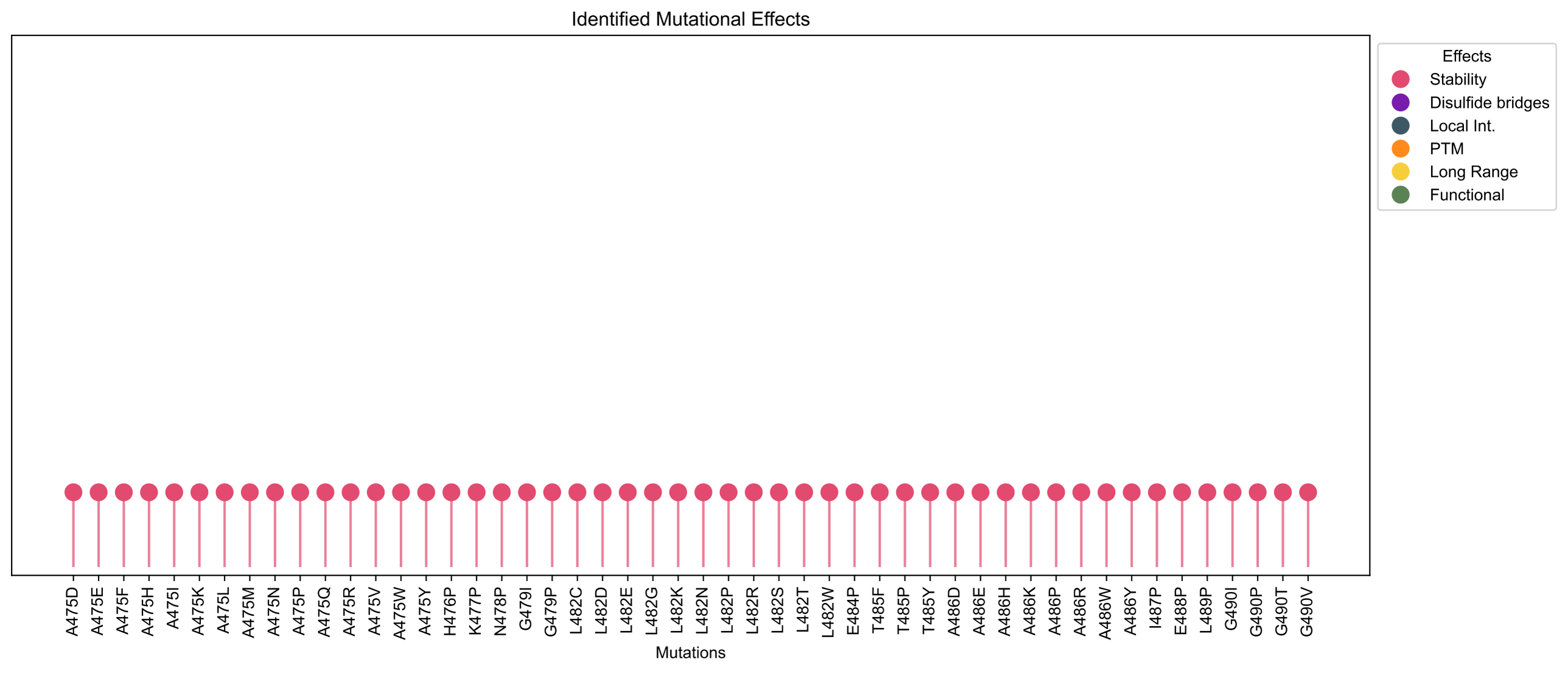

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
M412Y
I413C
I413D
I413E
I413G
I413H
I413K
I413N
I413P
I413Q
I413R
I413S
I413W
I413Y
K414P
N415P
V416D
V416E
V416G
V416H
V416K
V416P
V416R
V416S
V416W
V416Y
L417A
L417C
L417D
L417E
L417G
L417H
L417K
L417N
L417P
L417R
L417S
L417T
L417W
L417Y
H418P
S419F
S419H
S419P
S419W
S419Y
A420R
R421P
L422D
L422G
Mutations

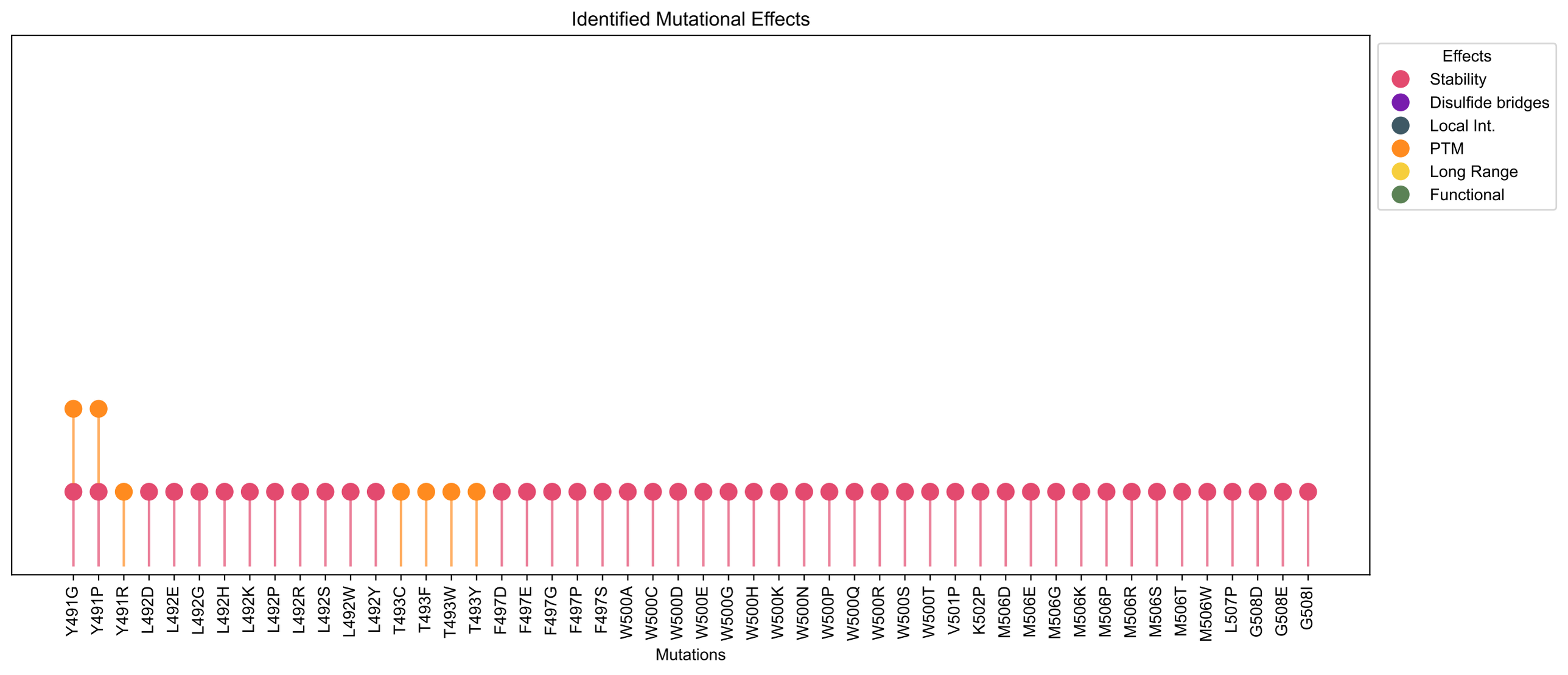

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
L422P
L423A
L423C
L423D
L423E
L423F
L423G
L423H
L423K
L423N
L423P
L423Q
L423R
L423S
L423T
L423W
L423Y
G424C
G424D
G424E
G424F
G424H
G424I
G424K
G424L
G424M
G424N
G424P
G424Q
G424R
G424V
G424W
G424Y
D425P
A426P
S427P
V428P
S429P
F430A
F430C
F430D
F430E
F430G
F430H
F430K
F430N
F430P
F430Q
F430R
F430S
Mutations

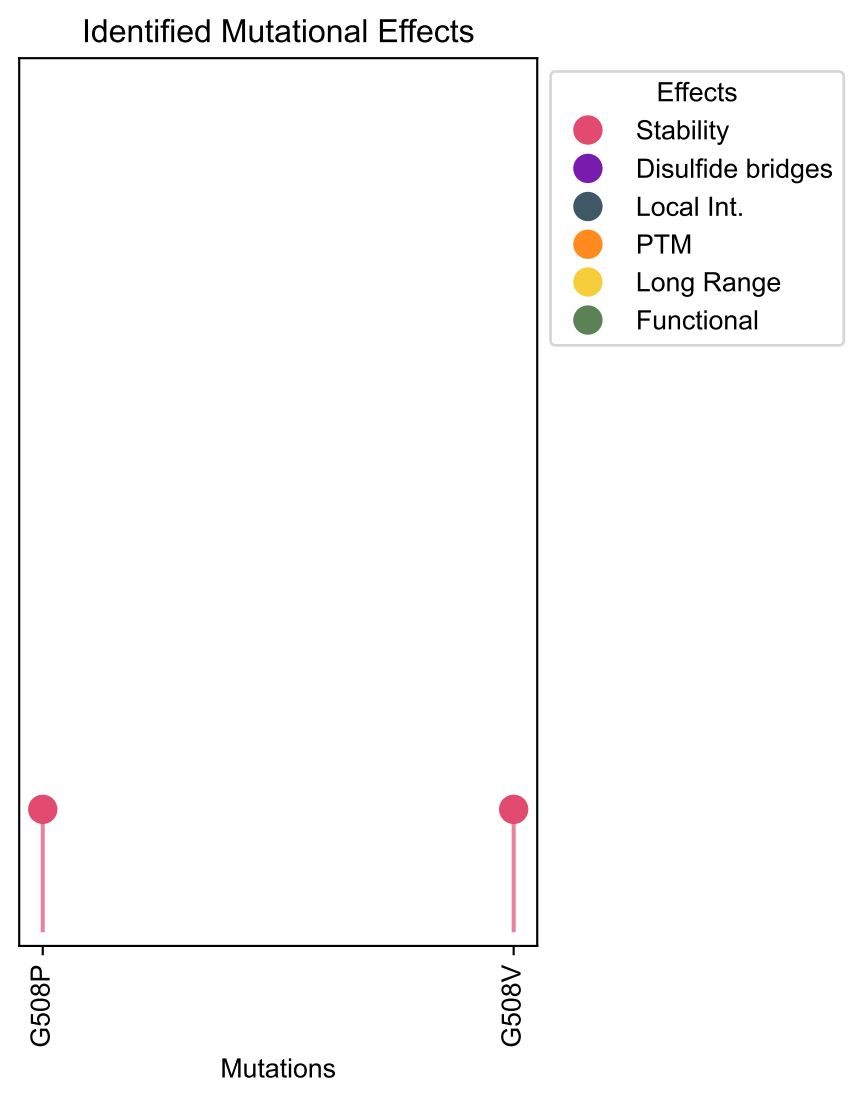

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
F430T
T431P
N433P
C434P
V435D
V435E
V435G
V435H
V435K
V435P
V435R
V435S
V435W
V435Y
V436P
G437P
I438A
I438C
I438D
I438E
I438G
I438H
I438K
I438N
I438P
I438Q
I438R
I438S
I438W
I438Y
Q439P
A440H
A440K
A440R
A440W
N441P
R444P
I445C
I445D
I445E
I445F
I445G
I445H
I445K
I445P
I445R
I445S
I445T
I445W
I445Y
Mutations


Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
K447P
L448G
L448P
M449D
M449E
M449G
M449P
M449S
M449W
N450P
S452F
S452H
S452P
S452W
S452Y
L455A
L455D
L455E
L455G
L455P
L455R
L455S
L455T
L455W
L455Y
A458D
A458E
A458F
A458H
A458W
A458Y
L459A
L459C
L459D
L459E
L459G
L459K
L459N
L459P
L459R
L459S
L459T
N460P
H462P
I463D
I463G
I463P
I463S
I463W
I463Y
Mutations


Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
G464I
G464P
G464V
Y465A
Y465C
Y465D
Y465E
Y465F
Y465G
Y465H
Y465I
Y465K
Y465L
Y465M
Y465N
Y465P
Y465Q
Y465R
Y465V
Y465W
A468D
A468E
A468F
A468H
A468I
A468K
A468L
A468N
A468P
A468Q
A468R
A468W
A468Y
A469P
K470P
I471D
I471E
I471G
I471H
I471K
I471N
I471P
I471Q
I471R
I471S
I471W
I471Y
A472P
K473P
T474P
Mutations


Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
A475D
A475E
A475F
A475H
A475I
A475K
A475L
A475M
A475N
A475P
A475Q
A475R
A475V
A475W
A475Y
H476P
K477P
N478P
G479I
G479P
L482C
L482D
L482E
L482G
L482K
L482N
L482P
L482R
L482S
L482T
L482W
E484P
T485F
T485P
T485Y
A486D
A486E
A486H
A486K
A486P
A486R
A486W
A486Y
I487P
E488P
L489P
G490I
G490P
G490T
G490V
Mutations


Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
Y491G
Y491P
Y491R
L492D
L492E
L492G
L492H
L492K
L492P
L492R
L492S
L492W
L492Y
T493C
T493F
T493W
T493Y
F497D
F497E
F497G
F497P
F497S
W500A
W500C
W500D
W500E
W500G
W500H
W500K
W500N
W500P
W500Q
W500R
W500S
W500T
V501P
K502P
M506D
M506E
M506G
M506K
M506P
M506R
M506S
M506T
M506W
L507P
G508D
G508E
G508I
Mutations


Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
G508P
G508V
Mutations